Adaptive Observers for Medical Digital Twins

A framework for personalized medicine

Juan Sereno

University of Dubrovnik

October 26, 2026

A motivating clinical case [1]

Panel 1. A 52-year-old male with a history of stroke and hemiplegia is found confused in his prison cell.



Taken, and adapted with Gemini, from: [1] Laubenbacher, R., Mehrad, B., Shmulevich, I., & Trayanova, N. (2024). **Digital twins in medicine**. *Nature Computational Science*, 4(3), 184–191.

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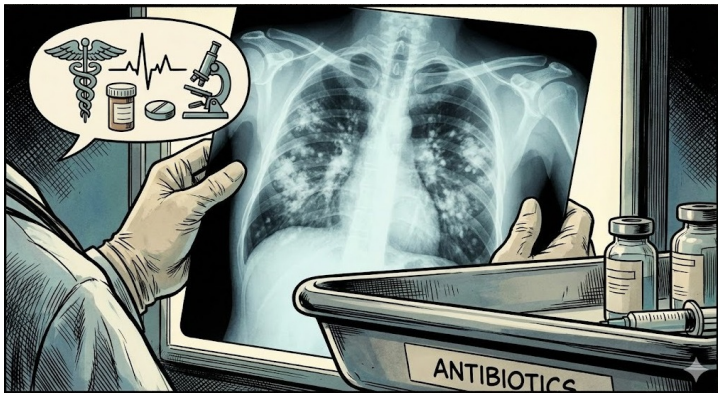
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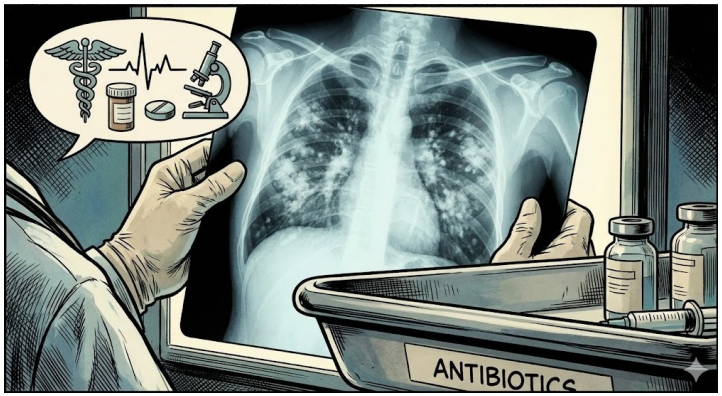
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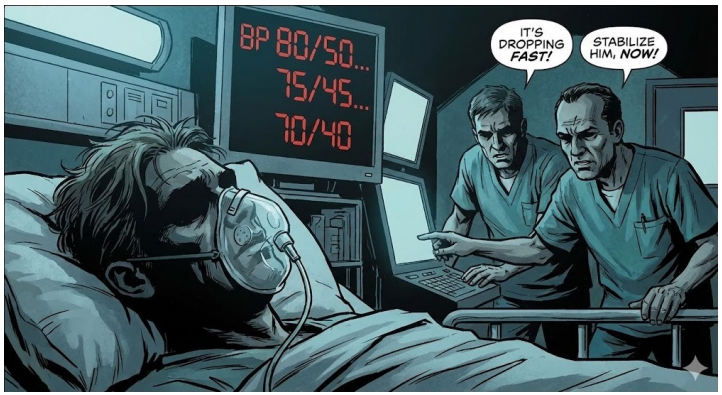
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Panel 4. 6 h later, recurrent hypotension requiring intravenous pressors and escalating supplemental oxygen.

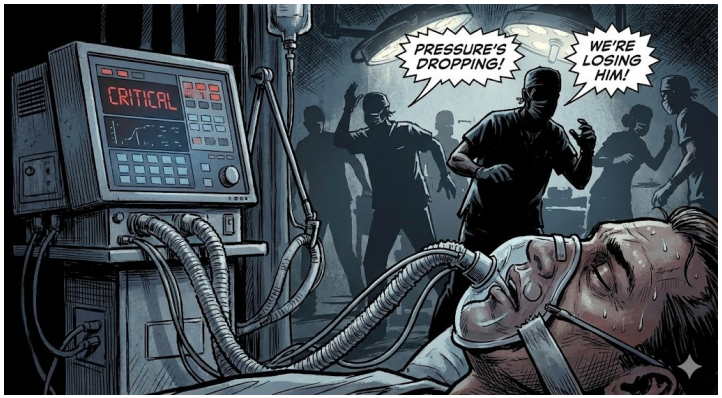


Suspected pathogen: *Klebsiella pneumoniae*.

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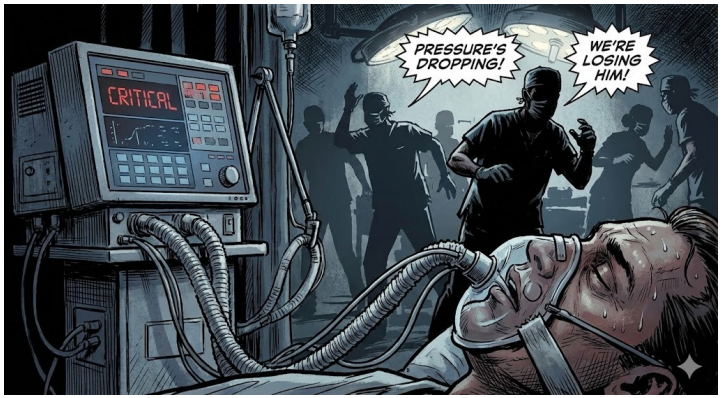
Panel 5. 3 h after deterioration, requires endotracheal intubation and mechanical ventilation for ARDS-induced hypoxic respiratory failure.



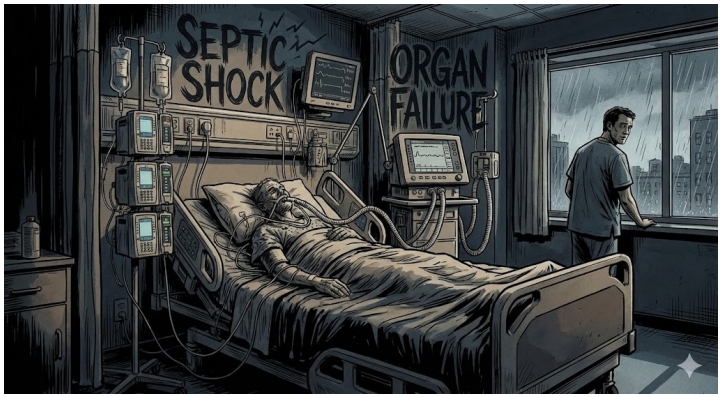
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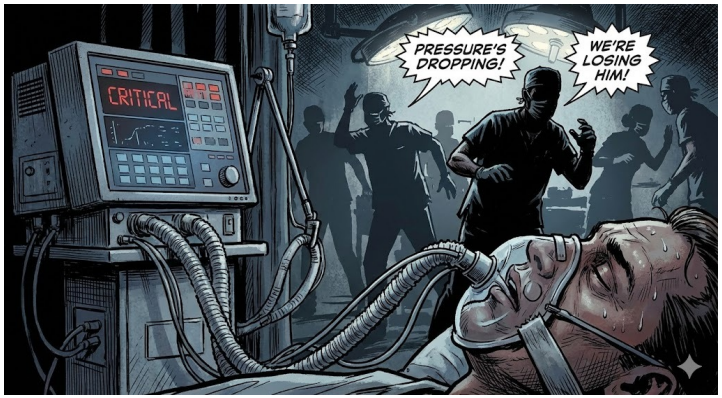
Panel 6. Condition progresses to refractory septic shock and acute kidney injury ***despite receiving antibiotics to which the K. pneumoniae was sensitive***.



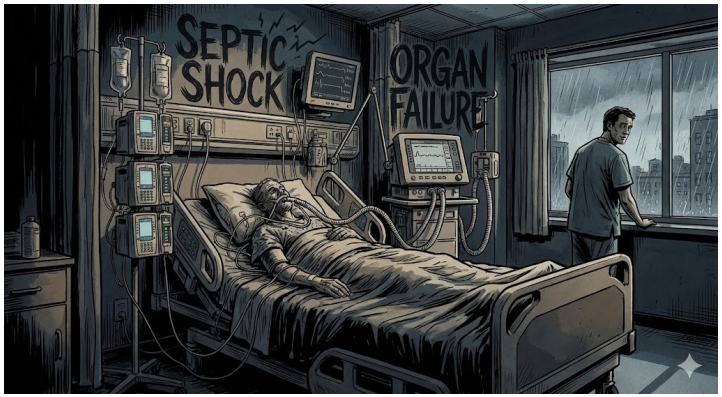
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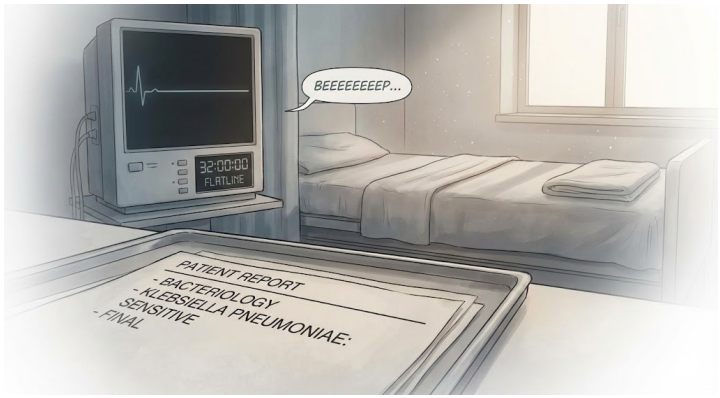
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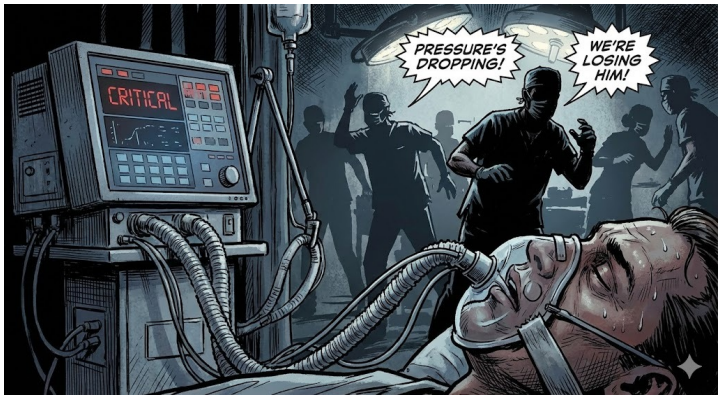
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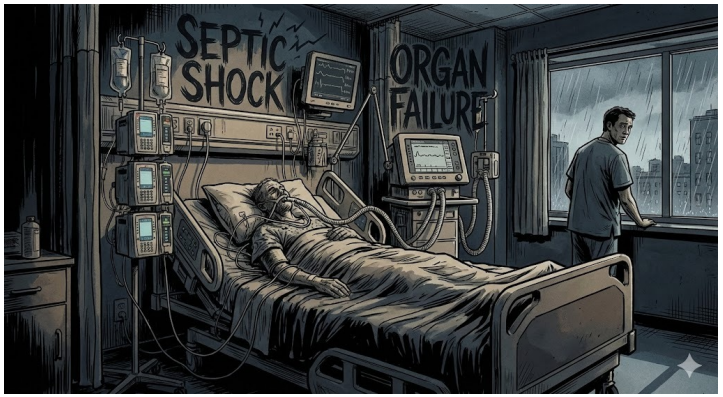
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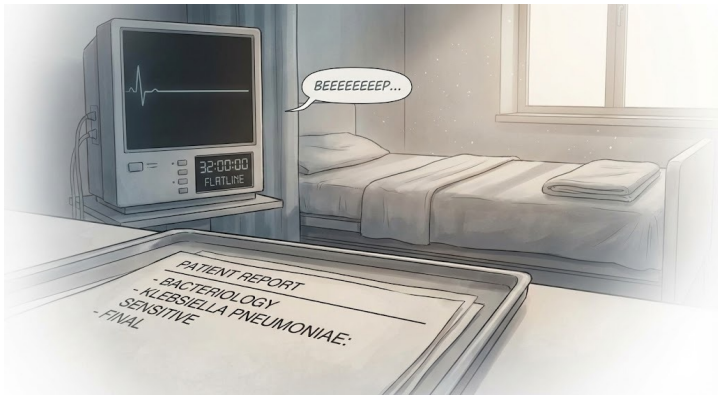
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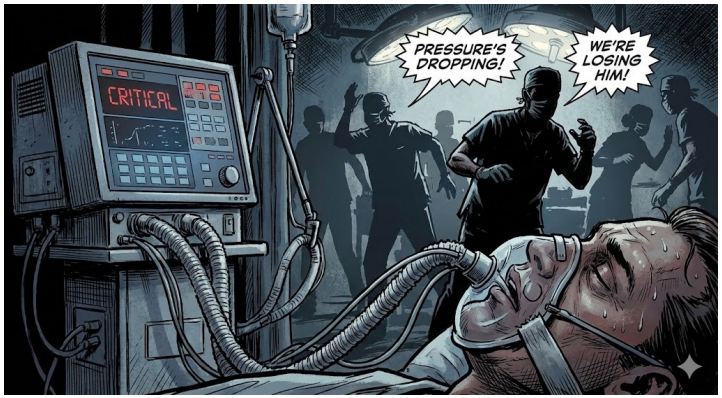
Panel 8. Real cases, closer to home. Jan 4, 2026.



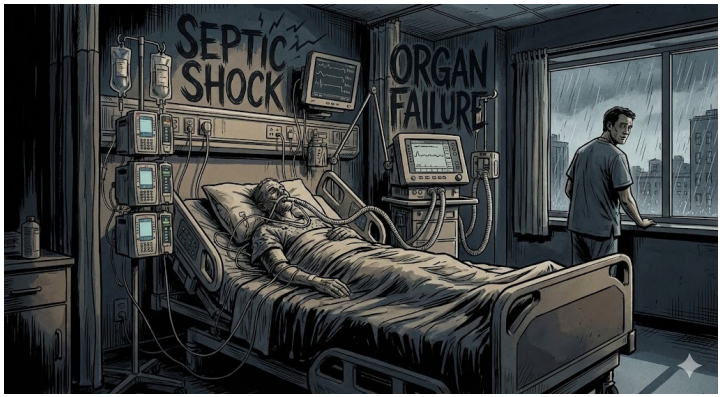
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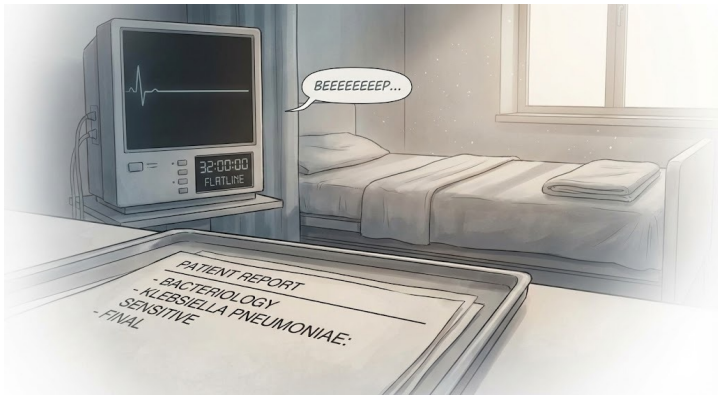
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Why do we need digital twins in medicine?

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What is a digital twin?

- A digital twin consists of three components [2]:
  1. A model of the physical twin;
  2. An evolving set of data relating to the object; and
  3. A means of dynamically updating or adjusting the model in accordance with the data.

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- The NASA digital twin is meant to be *ultra-realistic*, covering propulsion, energy storage, life support, avionics, thermal protection, etc.
- Between model fidelity and the ability to update the model from data, a trade-off arises.

[2] Wright, L., & Davidson, S. (2020). **How to tell the difference between a model and a digital twin**. *Adv. Model. Simul. Eng. Sci.*, 7(1), 13.
[3] Shafto, M., et al. (2012). **Modeling, Simulation, Information Technology and Processing Roadmap**. NASA, 32, 1–38.

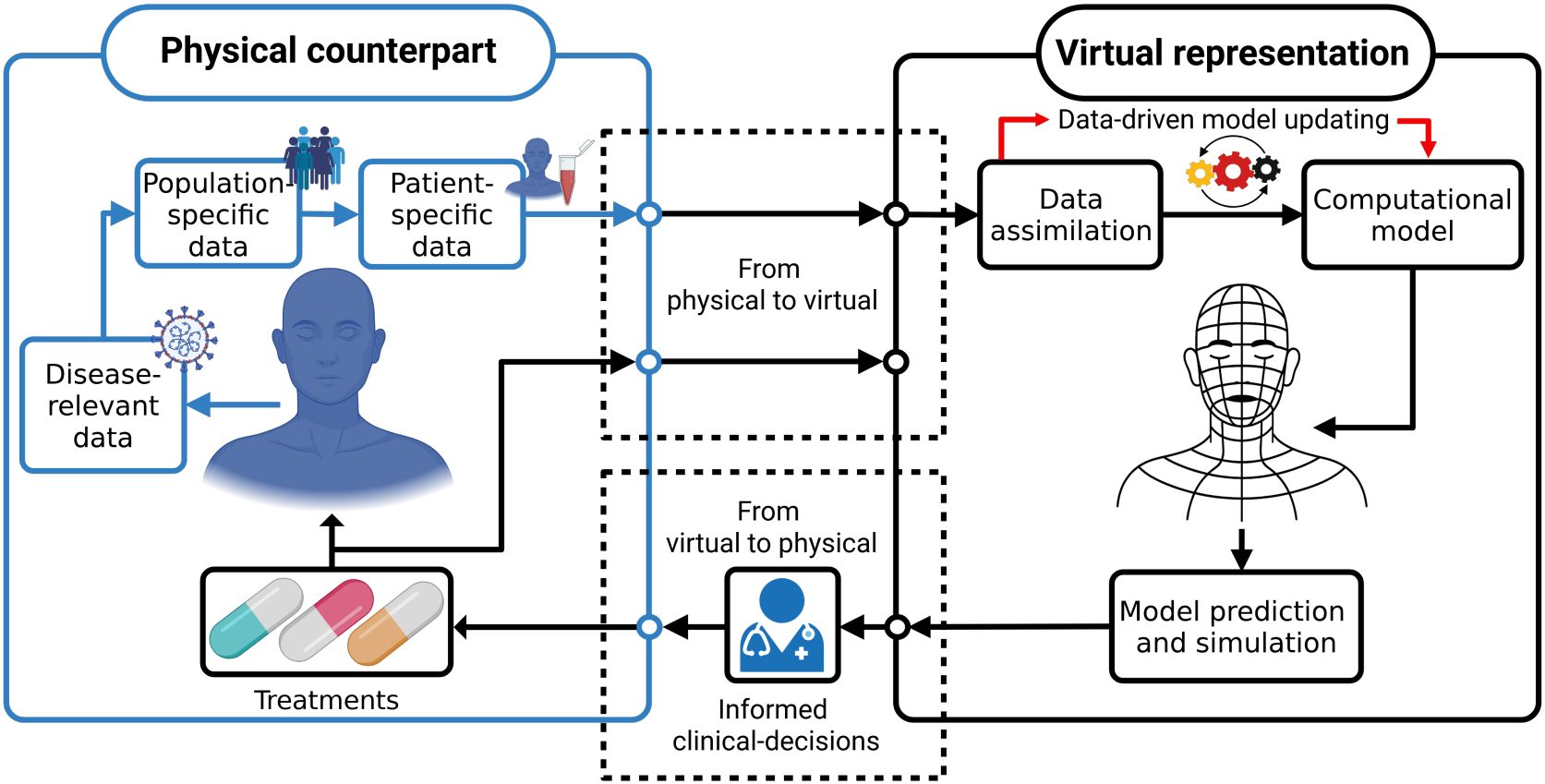
What is a medical digital twin?

- A medical digital twin [1,4] is a computational model of the system to be twinned (healthy or disease processes) that is
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  - connected to the physical twin in a bidirectional fashion over time,
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- Of the three ingredients of a digital twin, it is the third — the means of updating the model in accordance with data — that this talk is about.

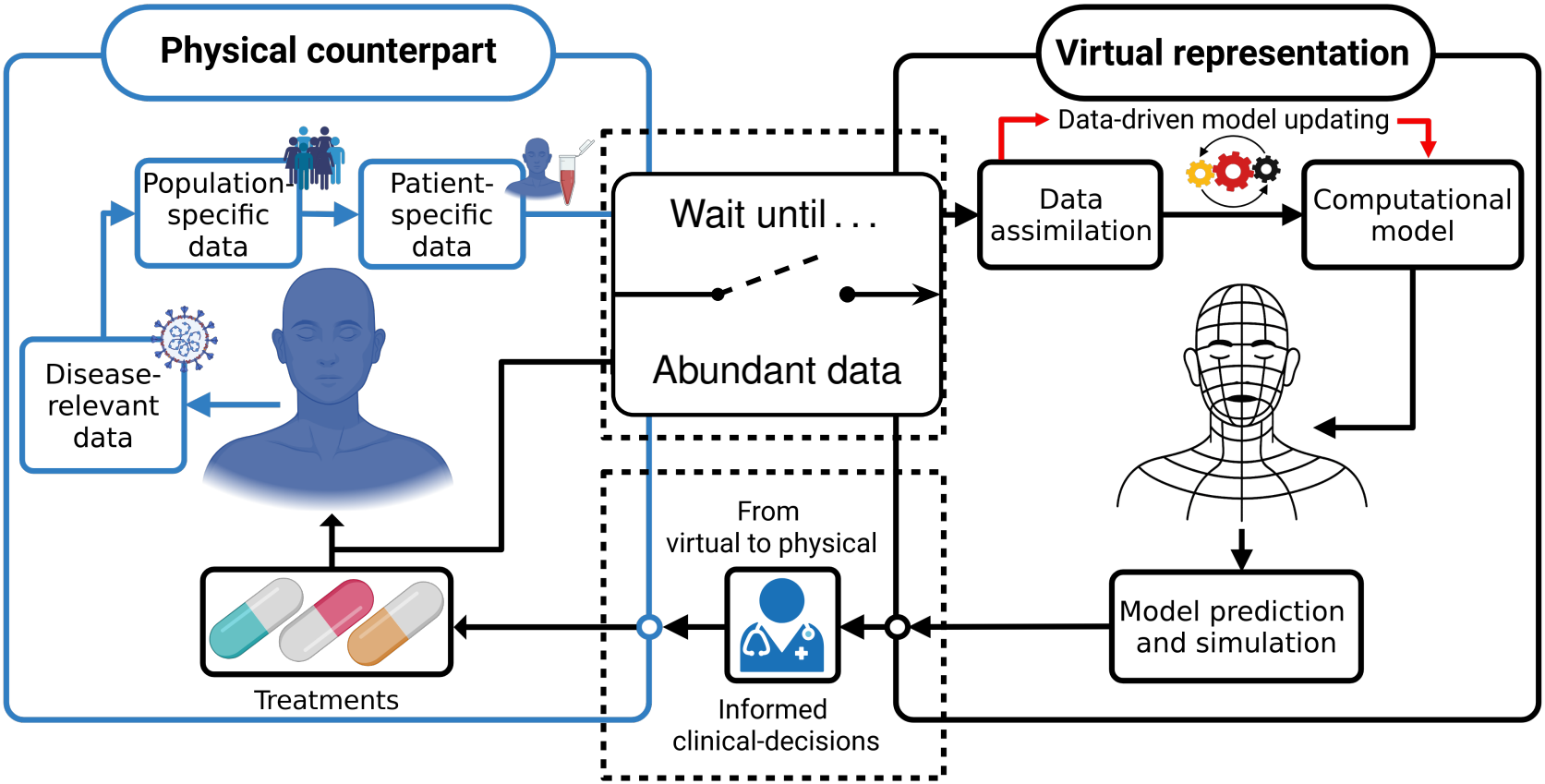
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[4] National Academies of Sciences, Engineering, and Medicine (2023). **Foundational Research Gaps and Future Directions for Digital Twins**. doi:10.17226/26894.



The standard digital twin framework [5].

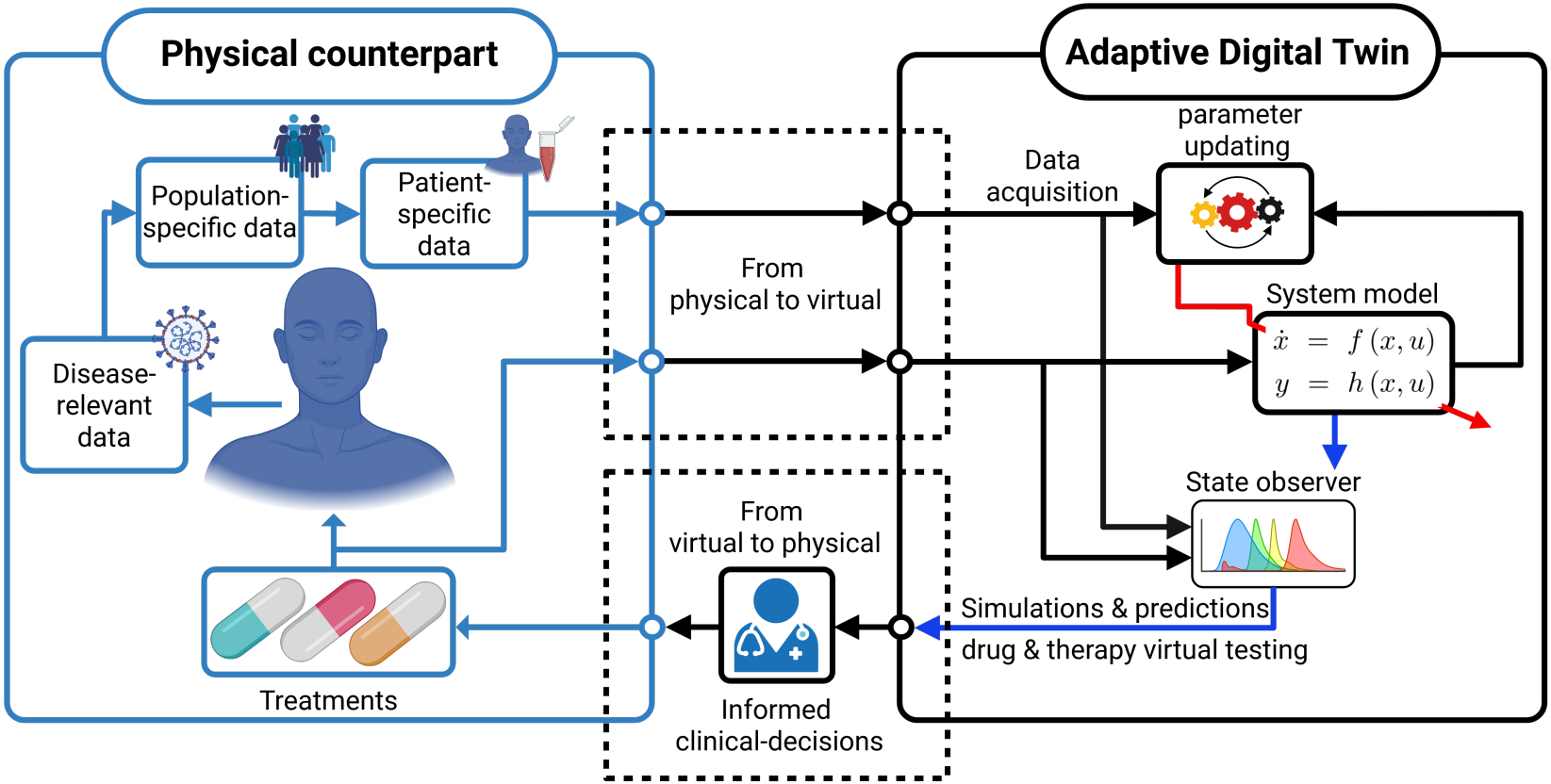
[5] Sereno, J. E., Neujahr, H., Hernandez-Gonzalez, M., & Hernandez-Vargas, E. A. (2025). **Iterative calibration of medical digital twins via adaptive estimators**. *Front. Appl. Math. Stat.*, 11, 1699390.

Traditional batch-based identification process



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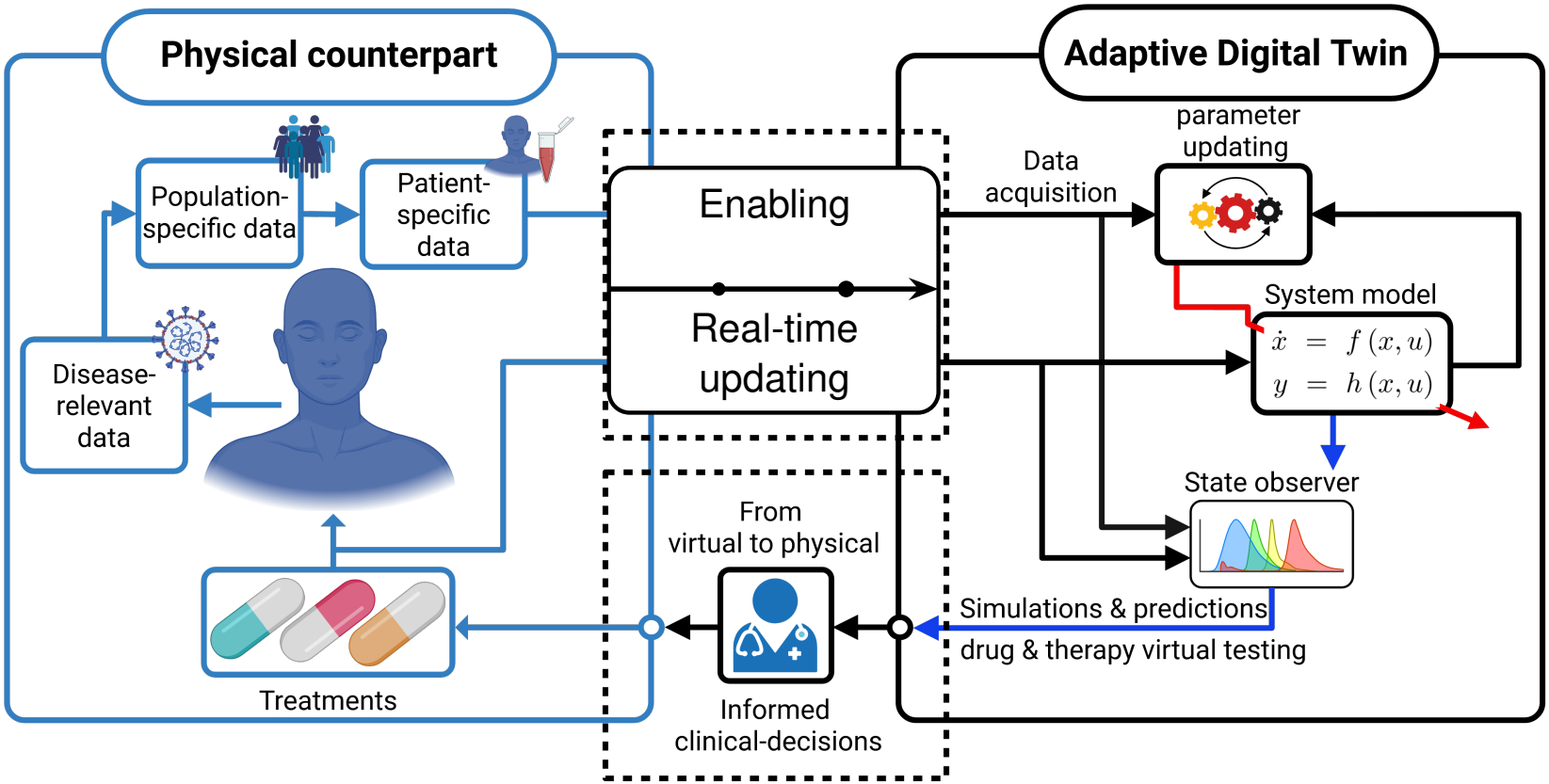
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Adaptive fashion: parameters are updated as new data arrive



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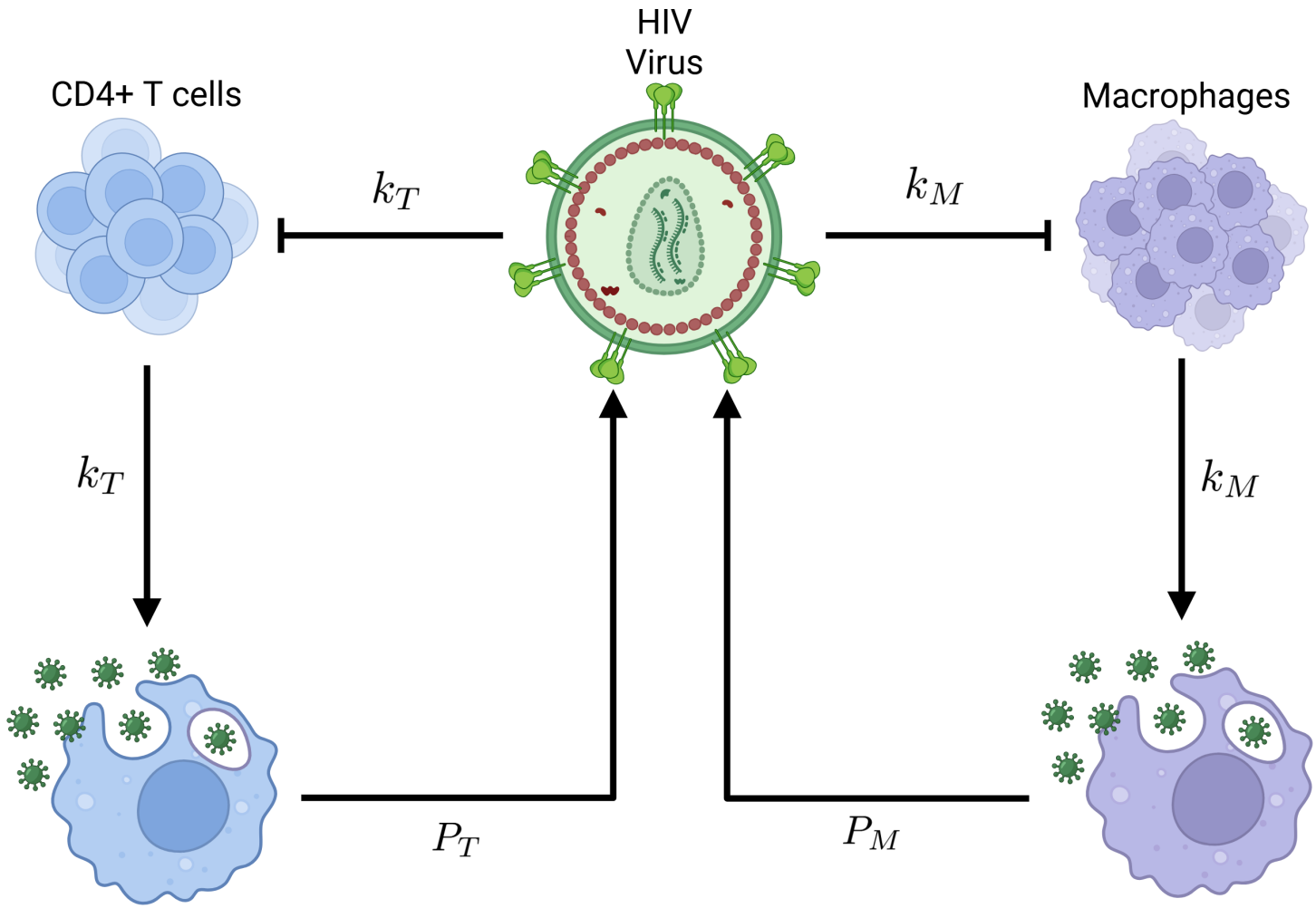
Mathematical foundations for model parameter updating

- Two tools for model parameter updating:
  1. Adaptive observers [6];
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- Biomedical application: an HIV mutation model [8].

[6] Besançon, G., de León-Morales, J., & Huerta-Guevara, O. (2006). **On adaptive observers for state affine systems**. *Int. J. Control*, 79(06), 581–591.
[7] Simon, D. (2006). **Optimal state estimation: Kalman, H infinity, and nonlinear approaches**. John Wiley & Sons.
[8] Hernandez-Vargas, E. A., & Middleton, R. H. (2013). **Modeling the three stages in HIV infection**. *J. Theor. Biol.*, 320, 33–40.



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The HIV mutation model [8]

$$
\begin{aligned}
\dot{T} &= s_T + \frac{\rho_T}{C_T + V_T} TV_T - \sum_{i=1}^{n} k^i_{T,\sigma} TV_i - \delta_T T, \\
\dot{M} &= s_M + \frac{\rho_M}{C_M + V_T} MV_T - \sum_{i=1}^{n} k^i_{M,\sigma} MV_i - \delta_M M, \\
\dot{T}^*_i &= k^i_{T,\sigma} TV_i + \sum_{j=1}^{n} \mu m_{i,j} V_j T - \delta_{T^*} T^*_i, \\
\dot{M}^*_i &= k^i_{M,\sigma} MV_i + \sum_{j=1}^{n} \mu m_{i,j} V_j M - \delta_{M^*} M^*_i, \\
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- Can we update model parameters (e.g. the infection rates $k^i_{T,\sigma}$) and estimate the number of infected cells from measurements?

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Adaptive observers

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- This approach assumes that the state estimation has faster dynamics than the parameter estimation.

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Adaptive observers [6]

- Consider a change of coordinates that rewrites model (1) in the state affine form

$$\dot{x} = A(y)x + \beta(y) + \varphi(y)\theta, \qquad y = Cx, \tag{2}$$

where $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$, $\theta \in \mathbb{R}^{n_\theta}$ are the state, the measured output and the unknown parameter vector.

- If y is measurable and $A(y)$, $\beta(y)$, $\varphi(y)$ are uniformly bounded continuous functions of y only, then the adaptive observer reads

$$\begin{aligned}
\dot{\hat{x}} &= A\hat{x} + \beta + \varphi\hat{\theta} + \left\{S^{-1}C^T + \Lambda\Gamma^{-1}\Lambda^T C^T\right\} Q\,(y - C\hat{x}),\\
\dot{S} &= -\rho_x S - A^T S - SA + C^T QC,\\
\dot{\Lambda} &= \left\{A - S^{-1}C^T QC\right\}\Lambda + \varphi,\\
\dot{\Gamma} &= -\rho_\theta \Gamma + \Lambda^T C^T QC\Lambda,\\
\dot{\hat{\theta}} &= \Gamma^{-1}\Lambda^T C^T Q\,(y - C\hat{x}).
\end{aligned} \tag{3}$$

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Adaptive observers for the HIV model

- For the HIV model (1) with one genotype, the change of coordinates

$$x = [T,\ M,\ T^*,\ M^*,\ V]/\lambda, \qquad y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} x$$

transforms (1) into

$$\begin{aligned}
\dot{x}_1 &= -\delta_T x_1 + \left(\frac{\lambda \rho_T y_1 y_2}{C_T + \lambda y_2} + \frac{s_T}{\lambda} \right) - \lambda y_1 y_2 K^1_{T,\sigma}, \\
\dot{x}_2 &= \left(\frac{\lambda \rho_M y_2}{C_M + \lambda y_2} - \lambda y_2 K^1_{M,\sigma} - \delta_M \right) x_2 + \frac{s_M}{\lambda}, \\
\dot{x}_3 &= -\delta_{T^*} x_3 + \lambda y_1 y_2 K^1_{T,\sigma}, \\
\dot{x}_4 &= \lambda y_2 K^1_{M,\sigma} x_2 - \delta_{M^*} x_4, \\
\dot{x}_5 &= P_{T,\sigma} x_3 + P_{M,\sigma} x_4 - \delta_V x_5,
\end{aligned}$$

where $\lambda \in \mathbb{R}$ is a known scaling factor, chosen for numerical convenience.

Adaptive observers for the HIV model

- This is of the state affine form (2), with

$$A = \begin{bmatrix} -\delta_T & 0 & 0 & 0 & 0 \\ 0 & \left(\frac{\lambda \rho_M y_2}{C_M + \lambda y_2} - \lambda y_2 K^1_{M,\sigma} - \delta_M\right) & 0 & 0 & 0 \\ 0 & 0 & -\delta_{T^*} & 0 & 0 \\ 0 & \lambda y_2 K^1_{M,\sigma} & 0 & -\delta_{M^*} & 0 \\ 0 & 0 & P_{T,\sigma} & P_{M,\sigma} & -\delta_V \end{bmatrix},$$

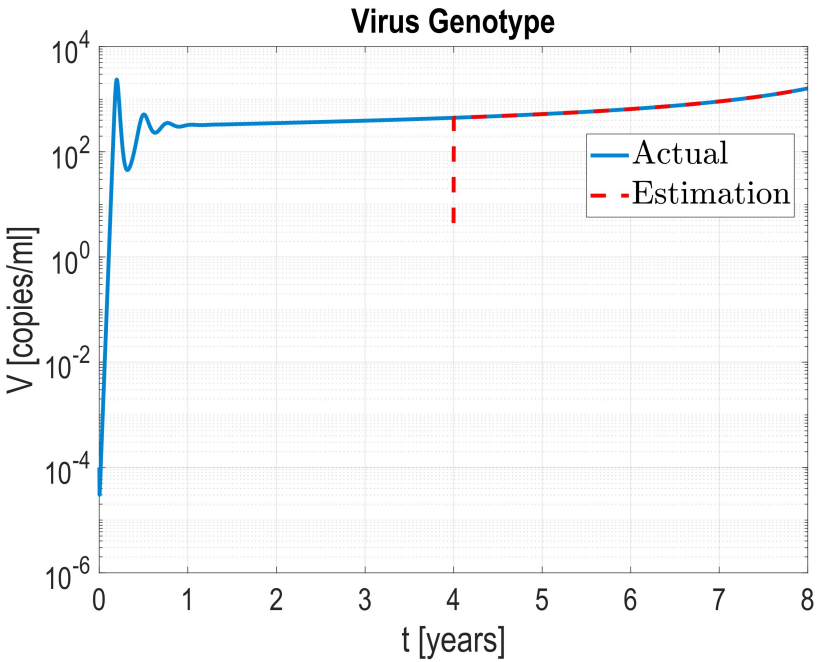
$$\beta = \begin{bmatrix} \frac{\lambda \rho_T y_1 y_2}{C_T + \lambda y_2} + \frac{s_T}{\lambda} \\ \frac{s_M}{\lambda} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \varphi = \begin{bmatrix} -\lambda y_1 y_2 \\ 0 \\ \lambda y_1 y_2 \\ 0 \\ 0 \end{bmatrix}, \qquad \theta = \begin{bmatrix} K^1_{T,\sigma} \end{bmatrix}.$$

- The unknown infection rate $K^1_{T,\sigma}$ becomes the parameter estimated online by the observer (3).

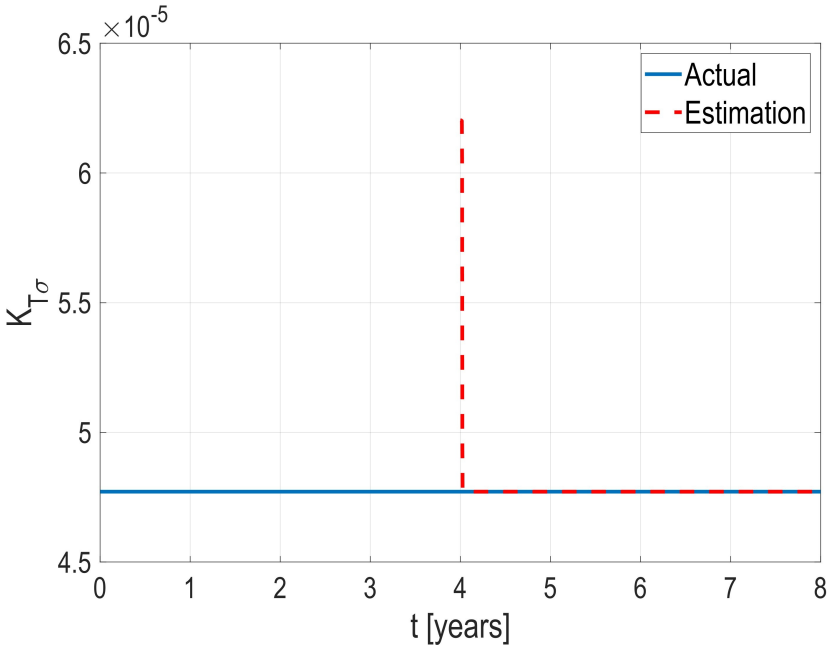
Results on real-time parameter updating

HIV model, one genotype

Scenario 1.1: one genotype, one parameter



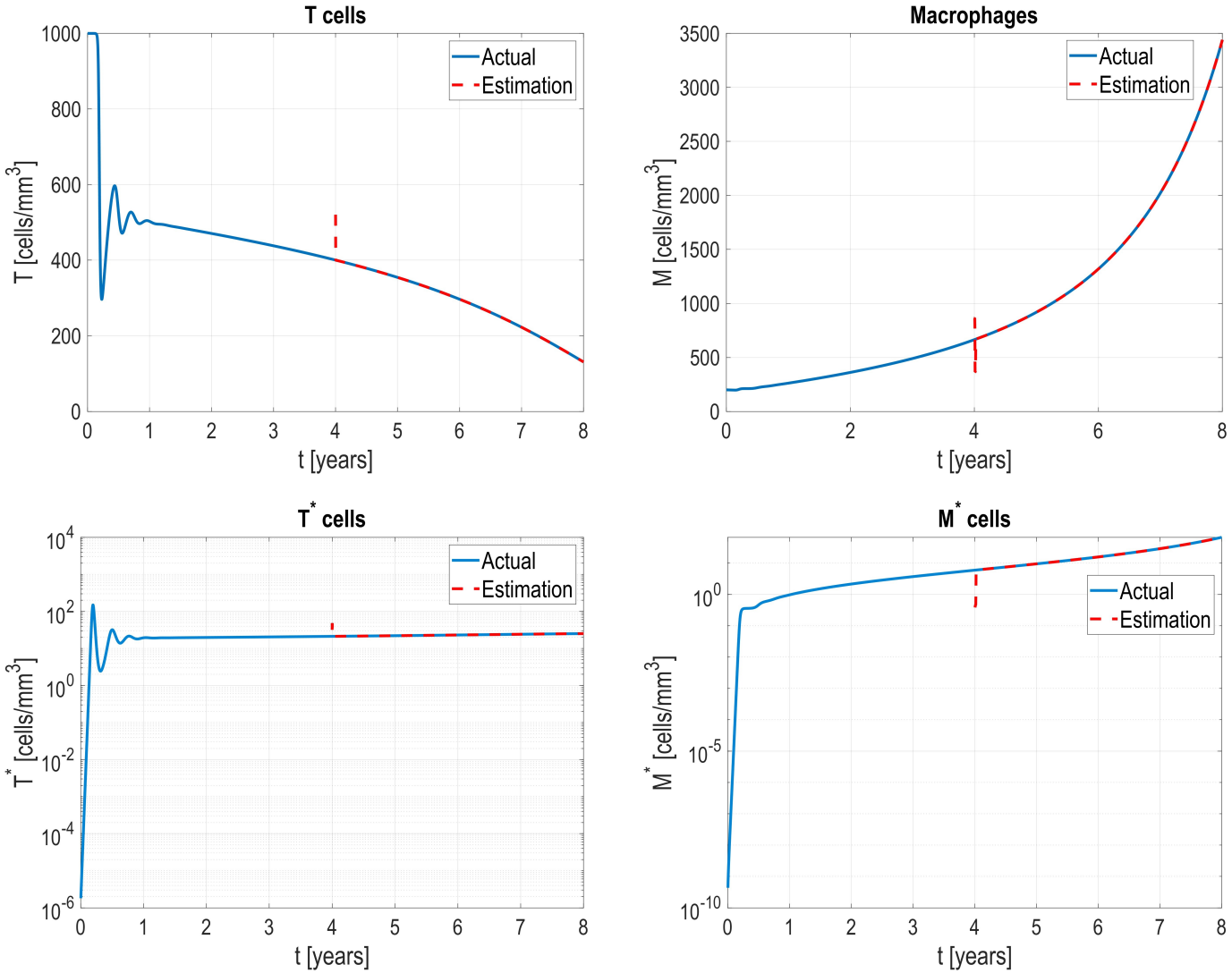
State estimation of the viral load.



Parameter estimation of $K_{T,\sigma}$.

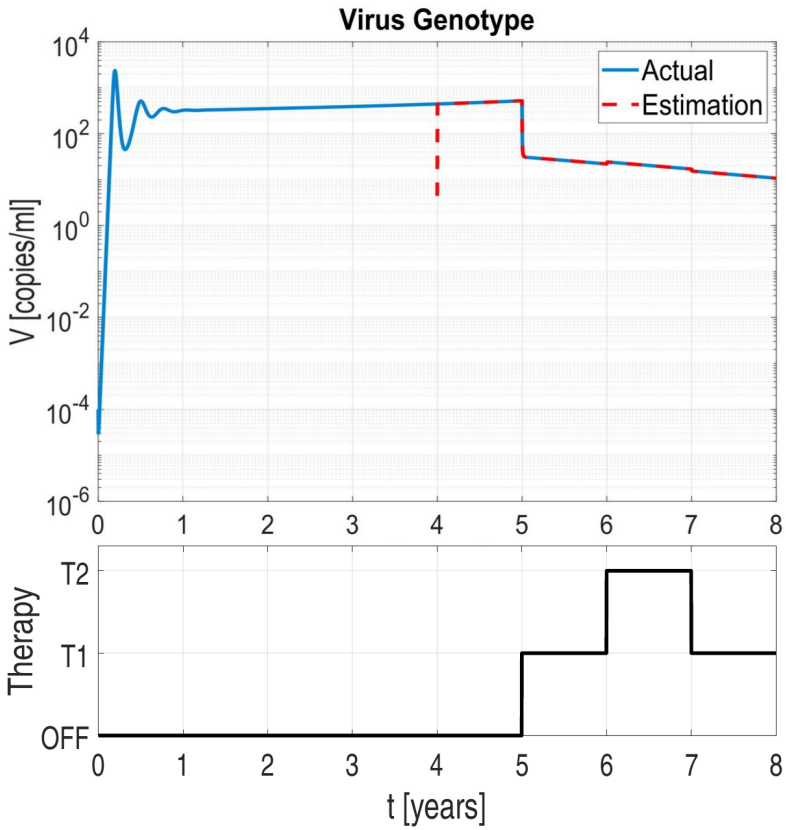
- The observer is switched on at $t = 4$ years; after a short transient the estimates lock on to the true trajectory and parameter.

Scenario 1.1: one genotype, one parameter

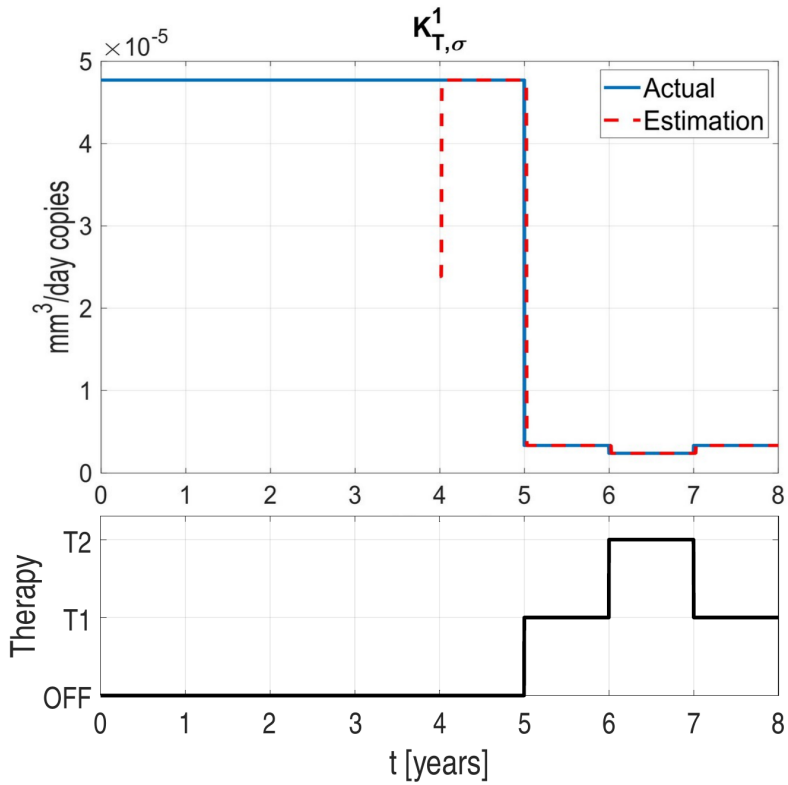


State estimation of the HIV model.

Scenario 1.2: switching between therapies

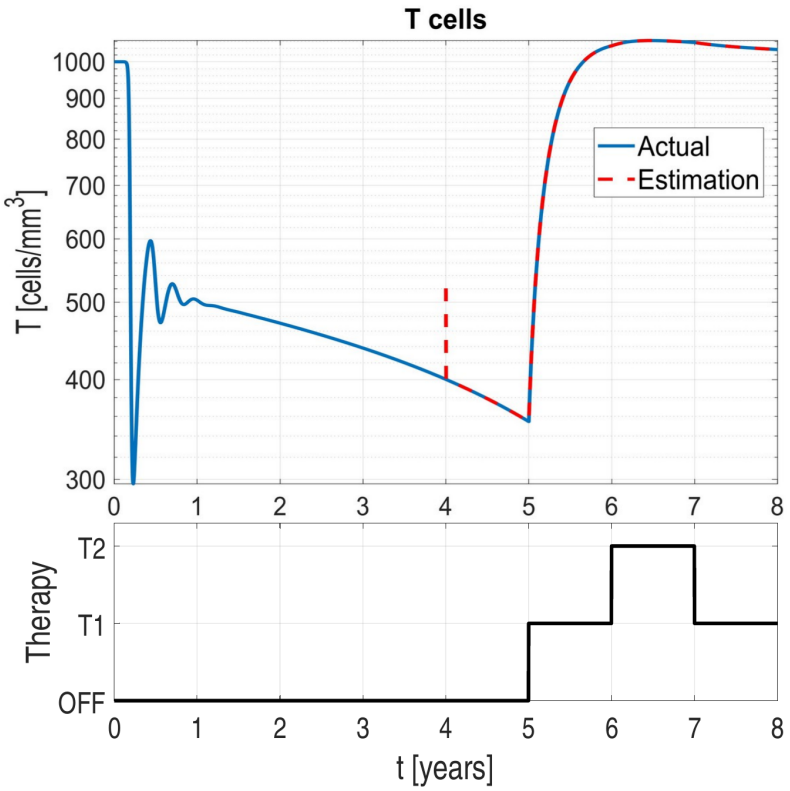


State estimation of the viral load.

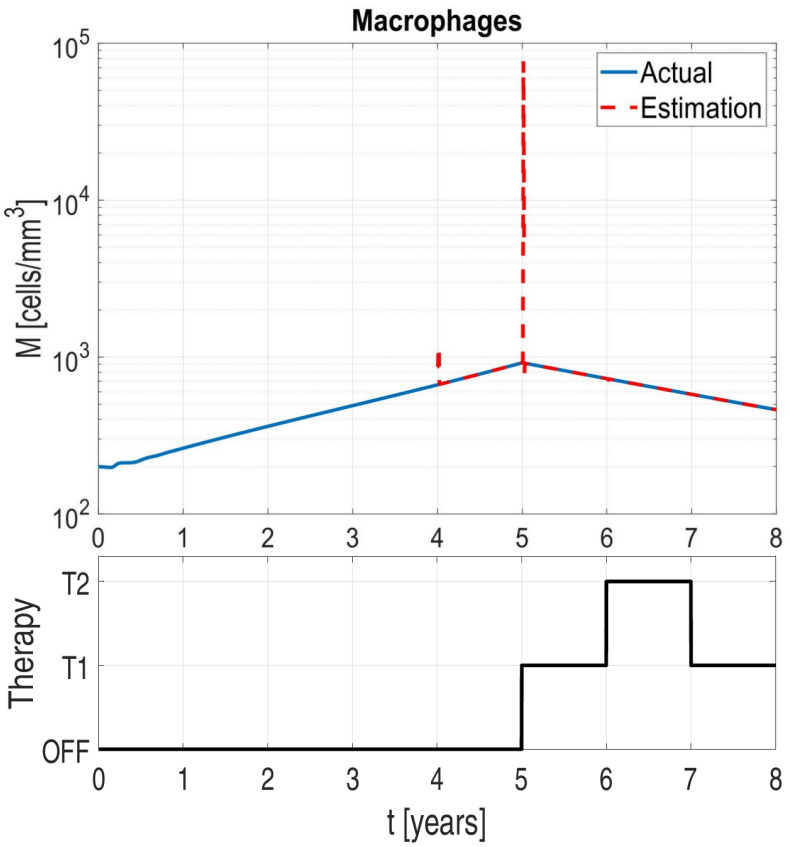


Parameter estimation of $K^1_{T,\sigma}$.

Scenario 1.2: switching between therapies

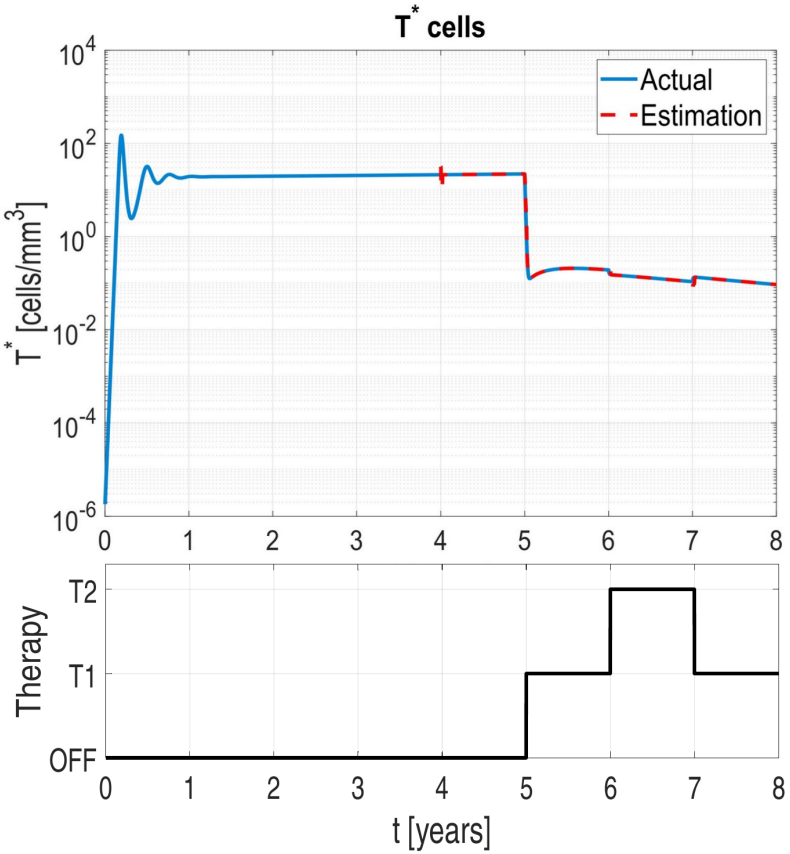


State estimation of CD4$^+$ T cells.

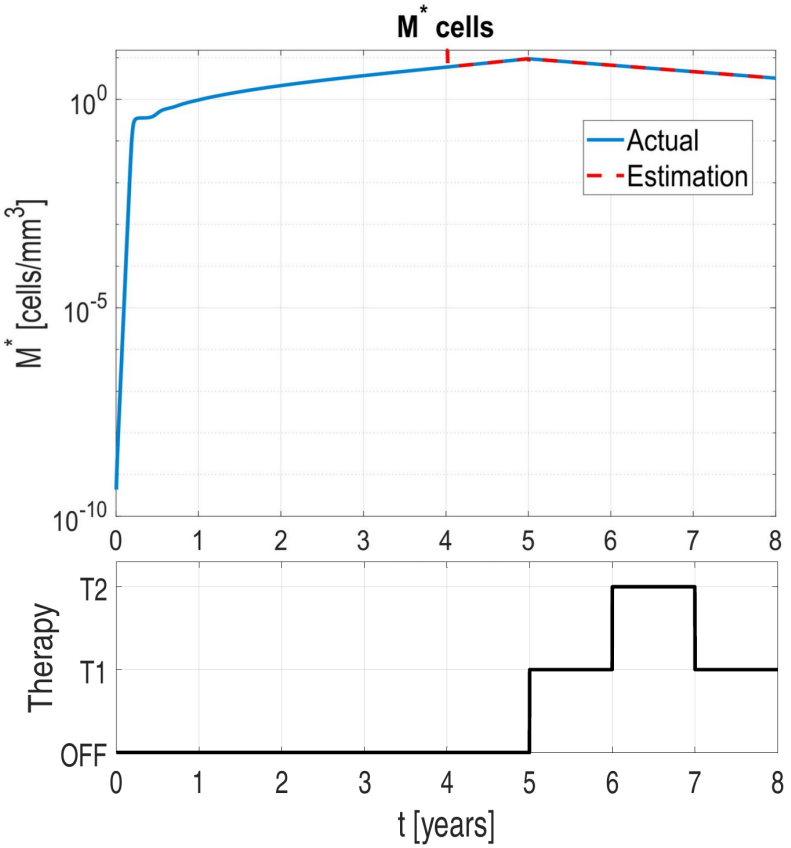


State estimation of macrophages.

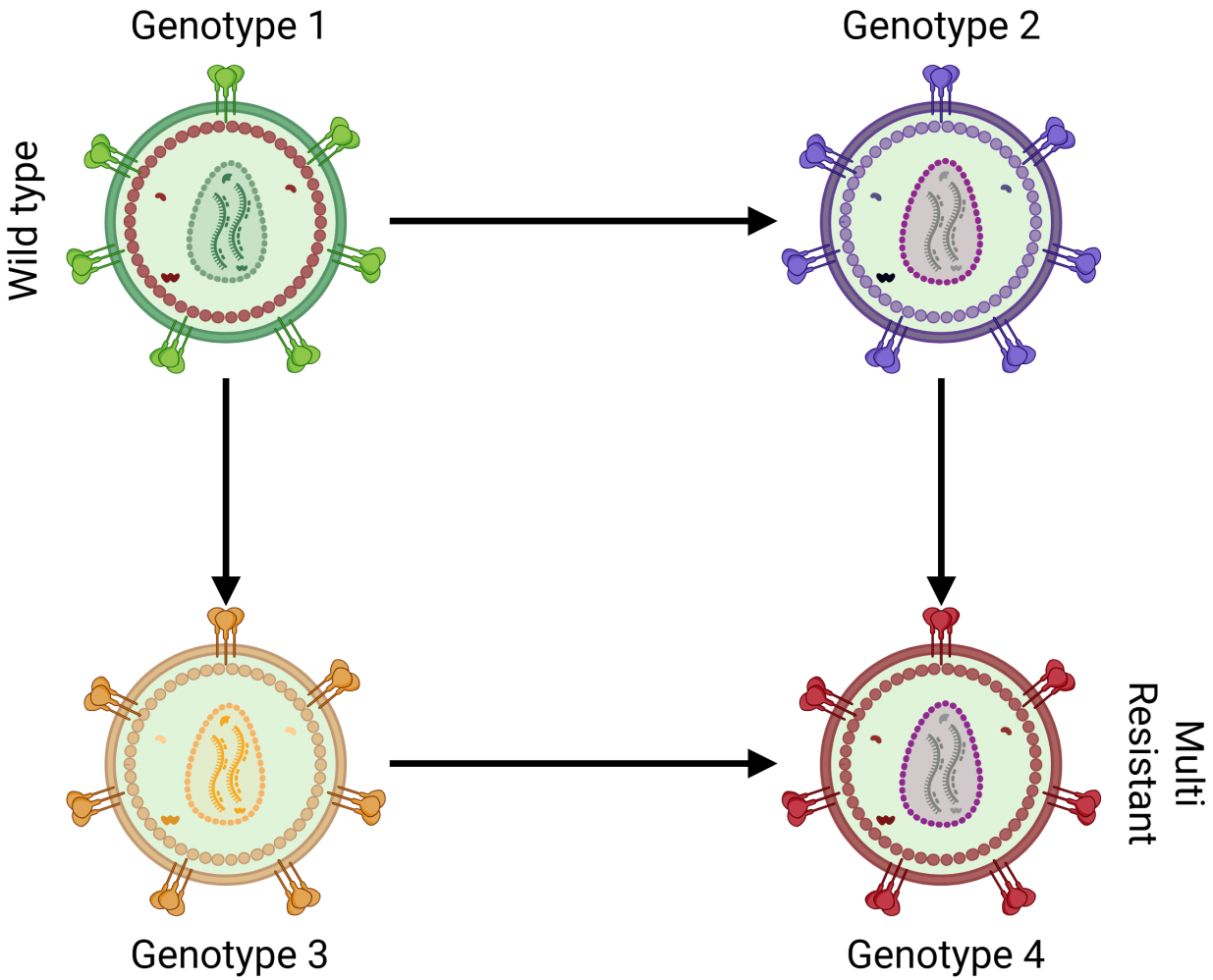
Scenario 1.2: switching between therapies



State estimation of infected T cells.



State estimation of infected macrophages.



HIV mutation model scheme with 4 genotypes [10].

[10] Hernandez-Vargas, E. A., Colaneri, P., & Middleton, R. H. (2013). **Switching strategies to mitigate HIV mutation**. *IEEE Trans. Control Syst. Technol.*, 22(4), 1623–1628.

HIV mutation: four genotypes, four parameters

- Consider the HIV model (1) with 4 genotypes and the change of coordinates

$$
\begin{aligned}
x = [\, & T,\ M,\ T_1^*,\ M_1^*,\ V_1,\ T_2^*,\ M_2^*,\ V_2, \\
& T_3^*,\ M_3^*,\ V_3,\ T_4^*,\ M_4^*,\ V_4\,]/\lambda, \\
y = [\, & T,\ V_1,\ V_2,\ V_3,\ V_4\,]/\lambda.
\end{aligned}
$$

HIV mutation: four genotypes, four parameters

- Consider the HIV model (1) with 4 genotypes and the change of coordinates

$$
\begin{aligned}
x = [\, & T,\ M,\ T_1^*,\ M_1^*,\ V_1,\ T_2^*,\ M_2^*,\ V_2, \\
& T_3^*,\ M_3^*,\ V_3,\ T_4^*,\ M_4^*,\ V_4\,]/\lambda, \\
y = [\, & T,\ V_1,\ V_2,\ V_3,\ V_4\,]/\lambda.
\end{aligned}
$$

- If $y_1 := T/\lambda$, $y_2 := V_1/\lambda$, $y_3 := V_2/\lambda$, $y_4 := V_3/\lambda$ and $y_5 := V_4/\lambda$ are measured for all time, the system can again be written in the state affine form (2).

HIV mutation: four genotypes, four parameters

- Consider the HIV model (1) with 4 genotypes and the change of coordinates

$$
\begin{aligned}
x = [\, & T,\ M,\ T_1^*,\ M_1^*,\ V_1,\ T_2^*,\ M_2^*,\ V_2, \\
& T_3^*,\ M_3^*,\ V_3,\ T_4^*,\ M_4^*,\ V_4\,]/\lambda, \\
y = [\, & T,\ V_1,\ V_2,\ V_3,\ V_4\,]/\lambda.
\end{aligned}
$$

- If $y_1 := T/\lambda$, $y_2 := V_1/\lambda$, $y_3 := V_2/\lambda$, $y_4 := V_3/\lambda$ and $y_5 := V_4/\lambda$ are measured for all time, the system can again be written in the state affine form (2).
- The matrices $A(y)$, $\beta(y)$, $\varphi(y)$ can then be computed, with the four unknown parameters

$$
\theta = \left[K_{T,\sigma}^1,\ K_{T,\sigma}^2,\ K_{T,\sigma}^3,\ K_{T,\sigma}^4\right]^T.
$$

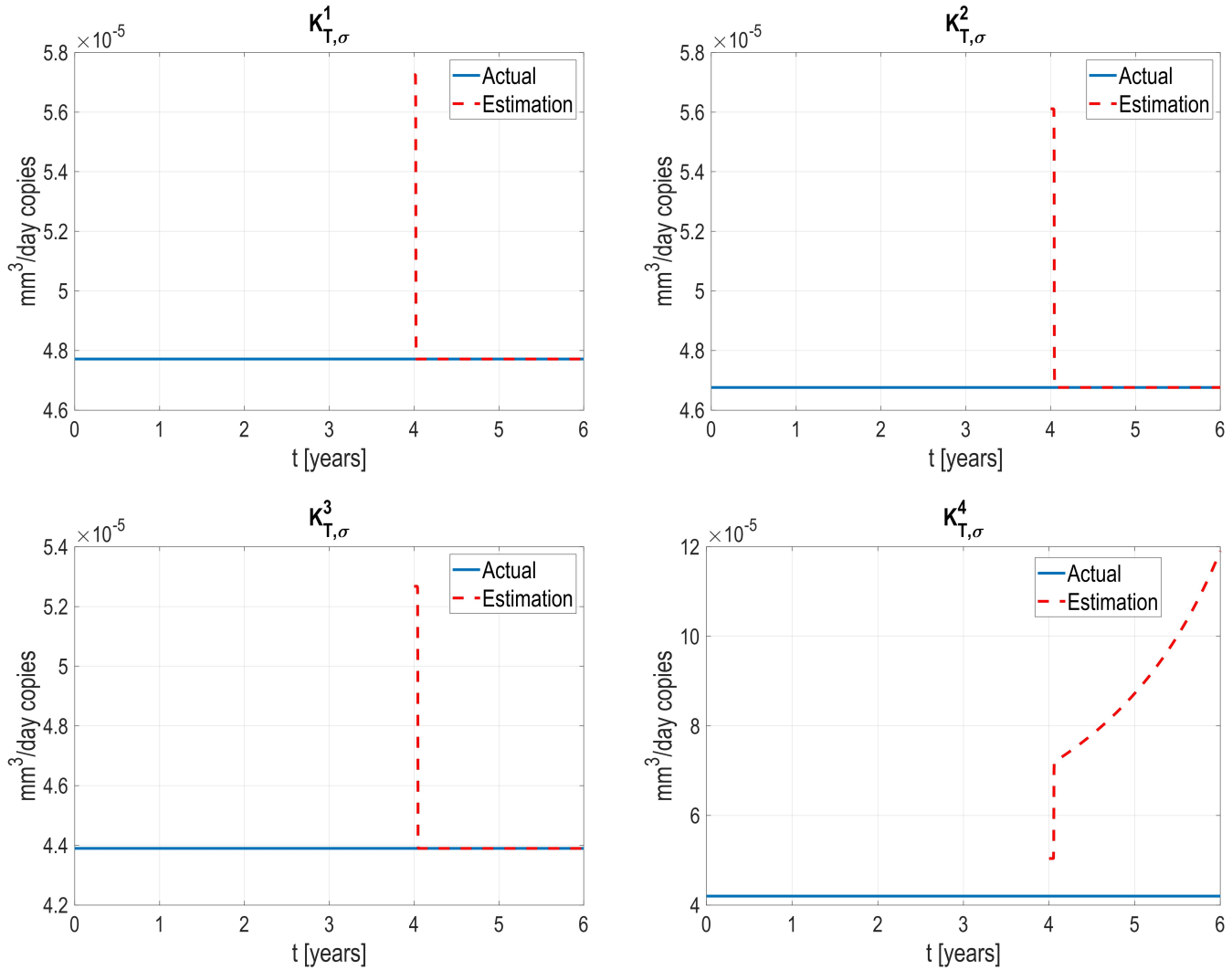
HIV mutation: four genotypes, four parameters

$$
\begin{aligned}
\dot{x}_1 &= -\delta_T x_1 + \left(\frac{\lambda \rho_T y_1 y_{V_T}}{C_T + \lambda y_{V_T}} + \frac{s_T}{\lambda}\right) - \lambda y_1 y_2 K^1_{T,\sigma} - \lambda y_1 y_3 K^2_{T,\sigma} - \lambda y_1 y_4 K^3_{T,\sigma} - \lambda y_1 y_5 K^4_{T,\sigma},\\
\dot{x}_2 &= \left(\frac{\lambda \rho_M y_{V_T}}{C_M + \lambda y_{V_T}} - \lambda y_2 K^1_{M,\sigma} - \lambda y_3 K^2_{M,\sigma} - \lambda y_4 K^3_{M,\sigma} - \lambda y_5 K^4_{M,\sigma} - \delta_M\right) x_2 + \frac{s_M}{\lambda},\\
\dot{x}_3 &= -\delta_{T^*} x_3 + (1-2\mu)\lambda y_1 y_2 K^1_{T,\sigma},\\
\dot{x}_4 &= (1-2\mu)\lambda y_2 K^1_{M,\sigma} x_2 - \delta_{M^*} x_4,\\
\dot{x}_5 &= P^1_{T,\sigma} x_3 + P^1_{M,\sigma} x_4 - \delta_V x_5,\\
\dot{x}_6 &= -\delta_{T^*} x_6 + \mu\lambda y_1 y_2 K^1_{T,\sigma} + (1-\mu)\lambda y_1 y_3 K^2_{T,\sigma},\\
\dot{x}_7 &= \left(\mu\lambda y_2 K^1_{M,\sigma} + (1-\mu)\lambda y_3 K^2_{M,\sigma}\right) x_2 - \delta_{M^*} x_7,\\
\dot{x}_8 &= P^2_{T,\sigma} x_6 + P^2_{M,\sigma} x_7 - \delta_V x_8,\\
\dot{x}_9 &= -\delta_{T^*} x_9 + \mu\lambda y_1 y_2 K^1_{T,\sigma} + (1-\mu)\lambda y_1 y_4 K^3_{T,\sigma},\\
\dot{x}_{10} &= \left(\mu\lambda y_2 K^1_{M,\sigma} + (1-\mu)\lambda y_4 K^3_{M,\sigma}\right) x_2 - \delta_{M^*} x_{10},\\
\dot{x}_{11} &= P^3_{T,\sigma} x_9 + P^3_{M,\sigma} x_{10} - \delta_V x_{11},\\
\dot{x}_{12} &= -\delta_{T^*} x_{12} + \mu\lambda y_1 y_3 K^2_{T,\sigma} + \mu\lambda y_1 y_4 K^3_{T,\sigma} + \lambda y_1 y_5 K^4_{T,\sigma},\\
\dot{x}_{13} &= \left(\mu\lambda y_3 K^2_{M,\sigma} + \mu\lambda y_4 K^3_{M,\sigma} + \lambda y_5 K^4_{M,\sigma}\right) x_2 - \delta_{M^*} x_{13},\\
\dot{x}_{14} &= P^4_{T,\sigma} x_{12} + P^4_{M,\sigma} x_{13} - \delta_V x_{14}.
\end{aligned}
$$

Results on real-time parameter updating

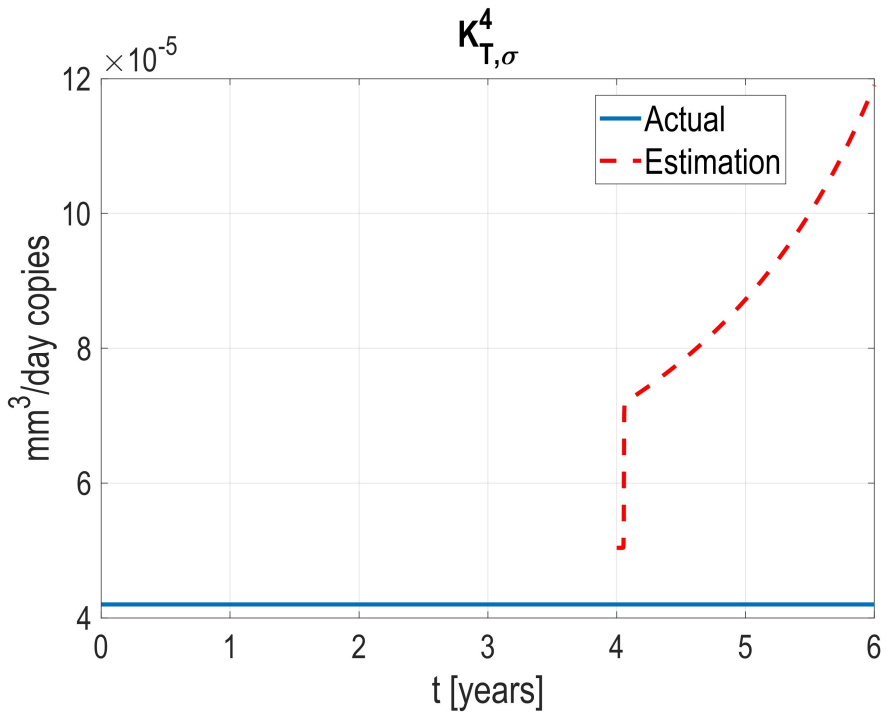
HIV mutation, four genotypes

Scenario 2.1: four genotypes, four parameters

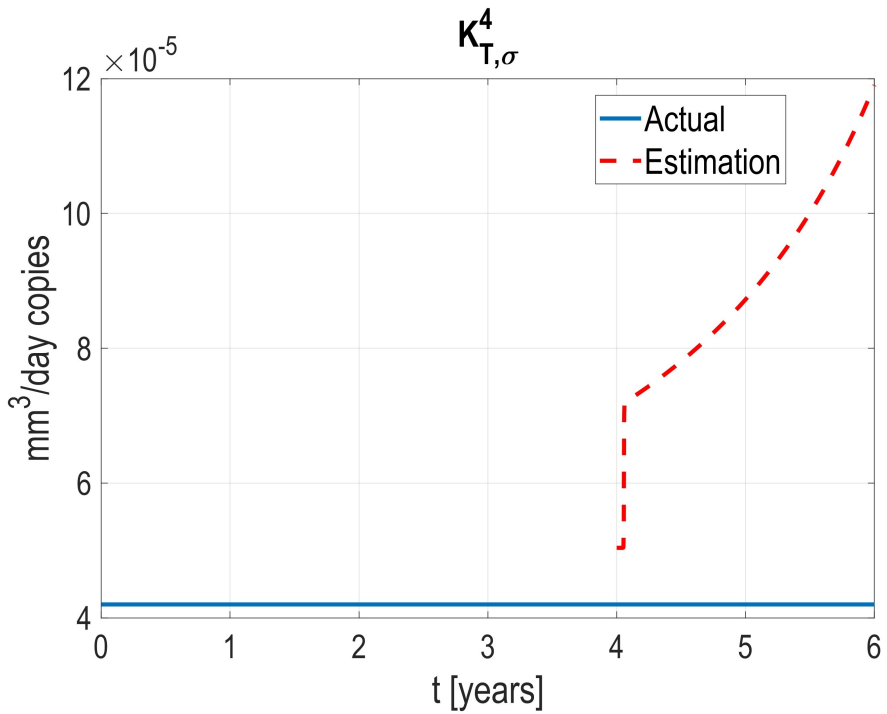


Parameter estimation of $K^i_{T,\sigma}$, $i = 1, 2, 3, 4$.

Scenario 2.1: four genotypes, four parameters

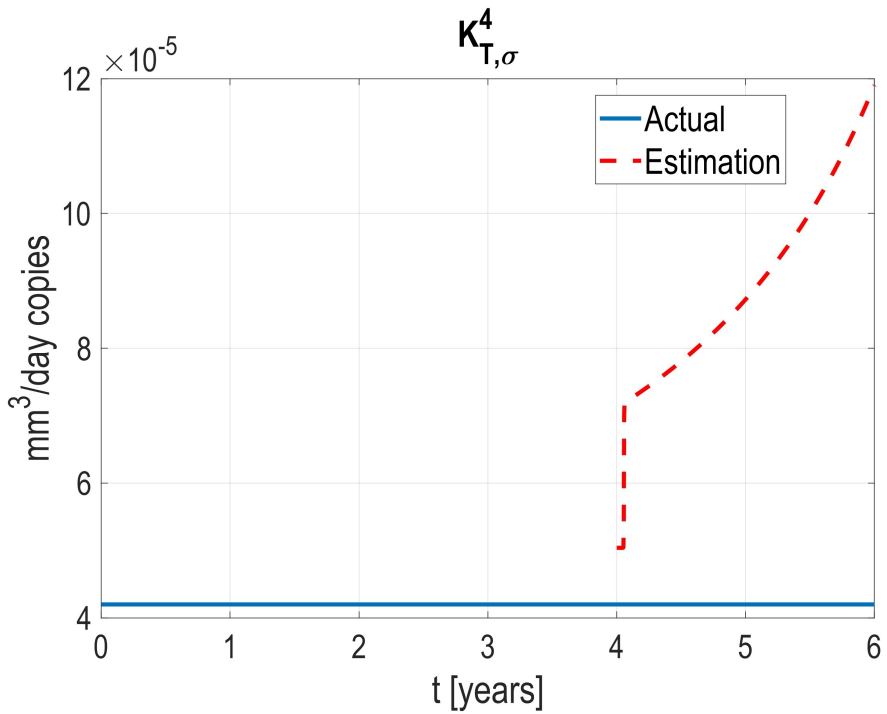


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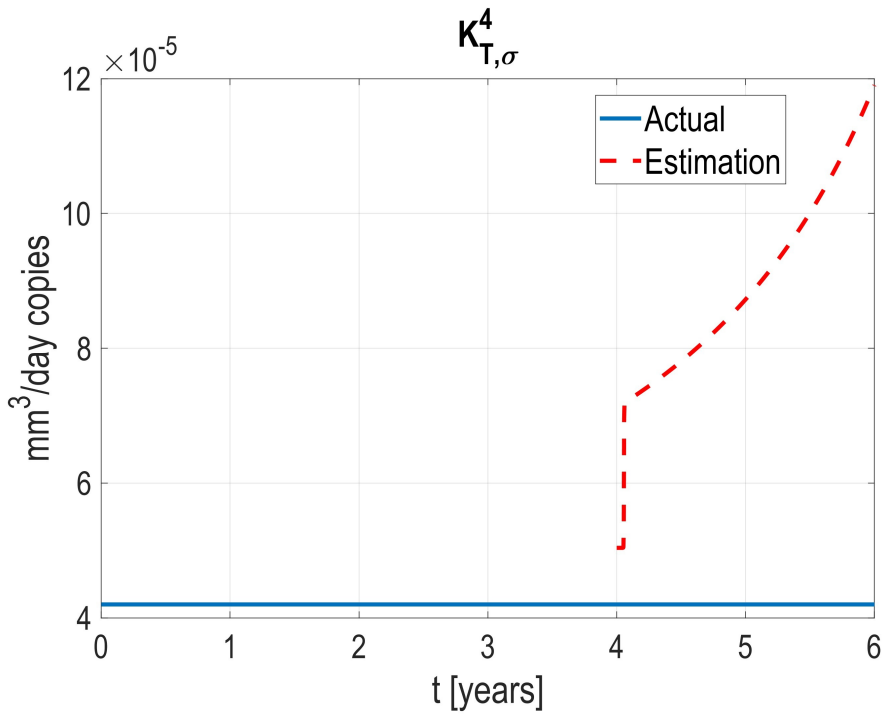
- The estimate of $K^4_{T,\sigma}$ does not converge.

Scenario 2.1: four genotypes, four parameters

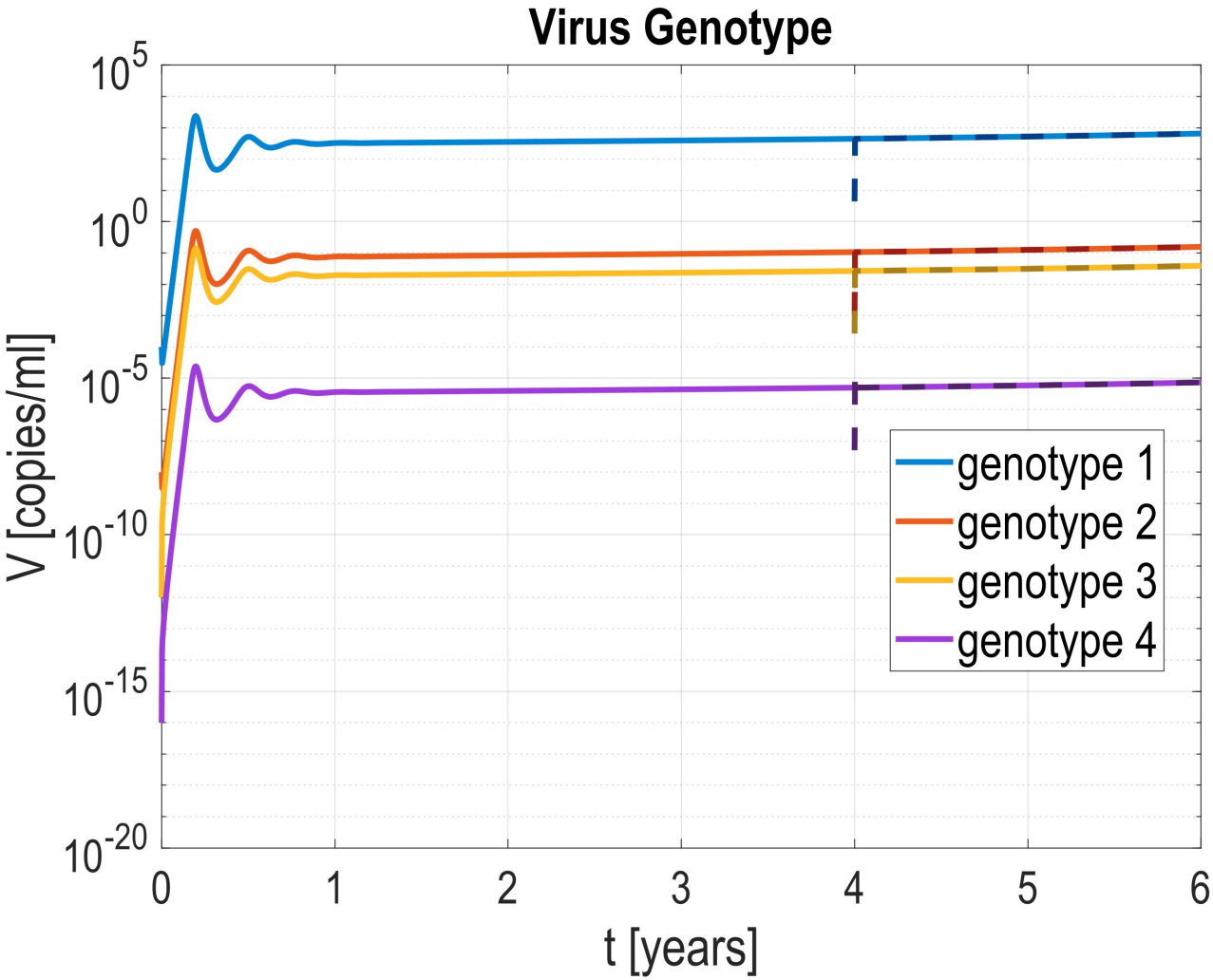


- The estimate of $K^4_{T,\sigma}$ does not converge.
- Genotype 4 (the multi-resistant strain) is not measurable in this setting.

Scenario 2.1: four genotypes, four parameters

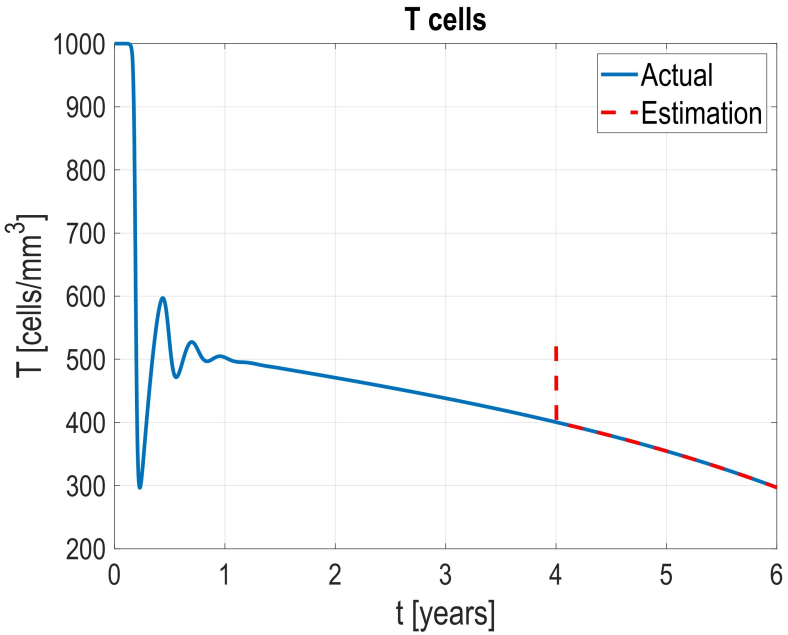


- The estimate of $K^4_{T,\sigma}$ does not converge.
- Genotype 4 (the multi-resistant strain) is not measurable in this setting.
- Therefore, $K^4_{T,\sigma}$ is not estimable.

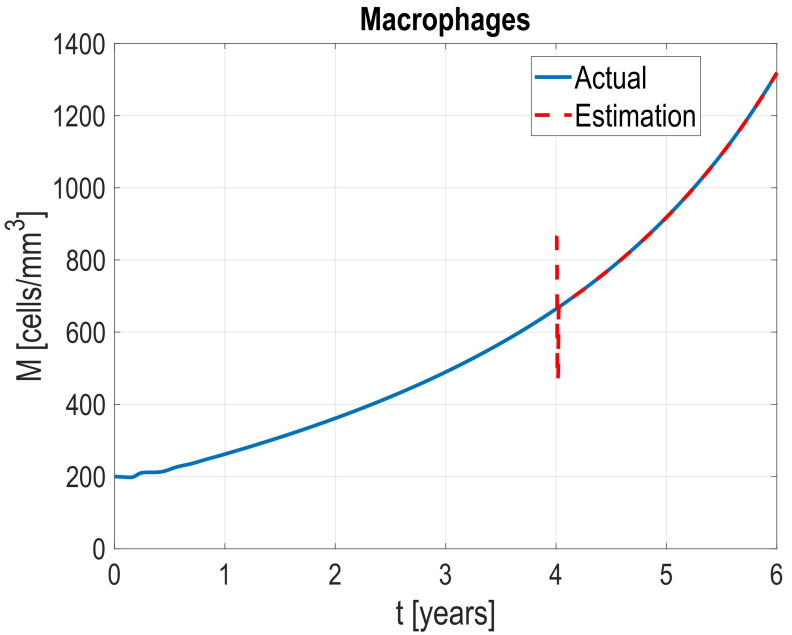


Scenario 2.1: state estimation of each virus genotype.

Scenario 2.1: four genotypes, four parameters

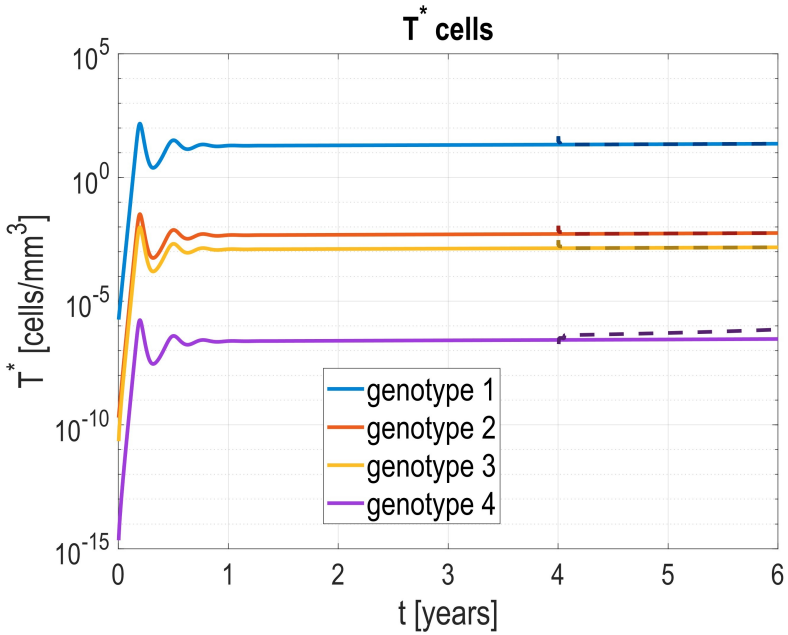


State estimation of CD4$^+$ T cells.

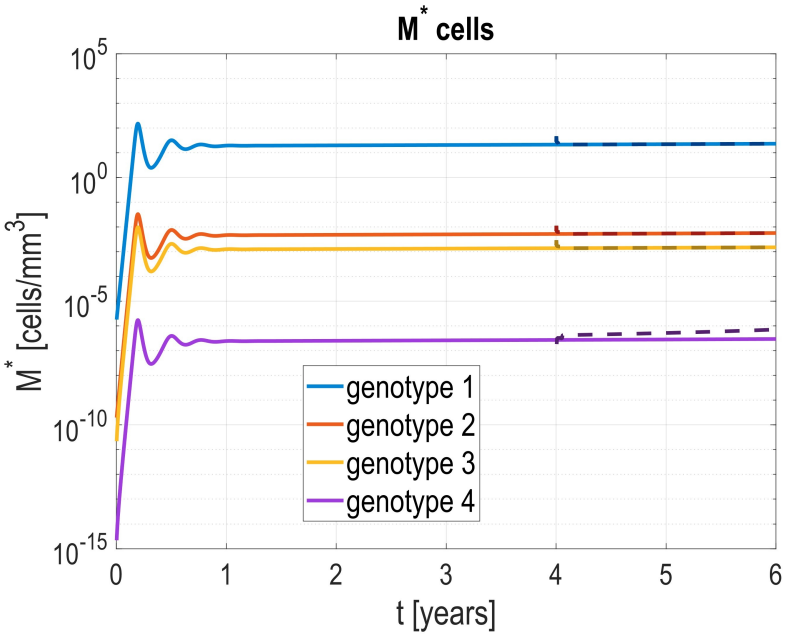


State estimation of macrophages.

Scenario 2.1: four genotypes, four parameters

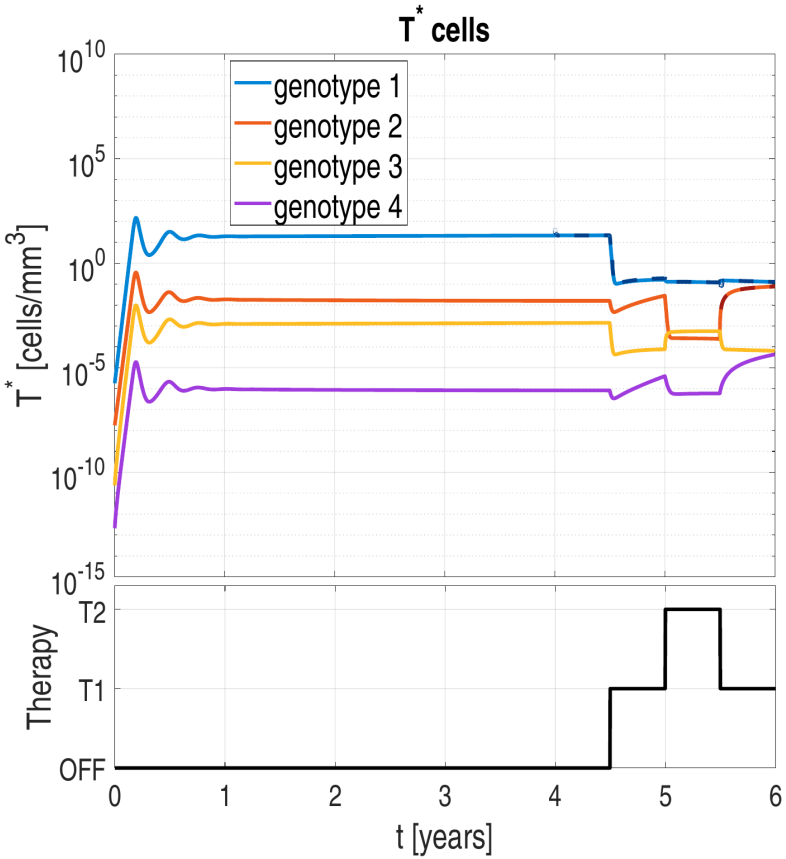


State estimation of infected T cells, by genotype.

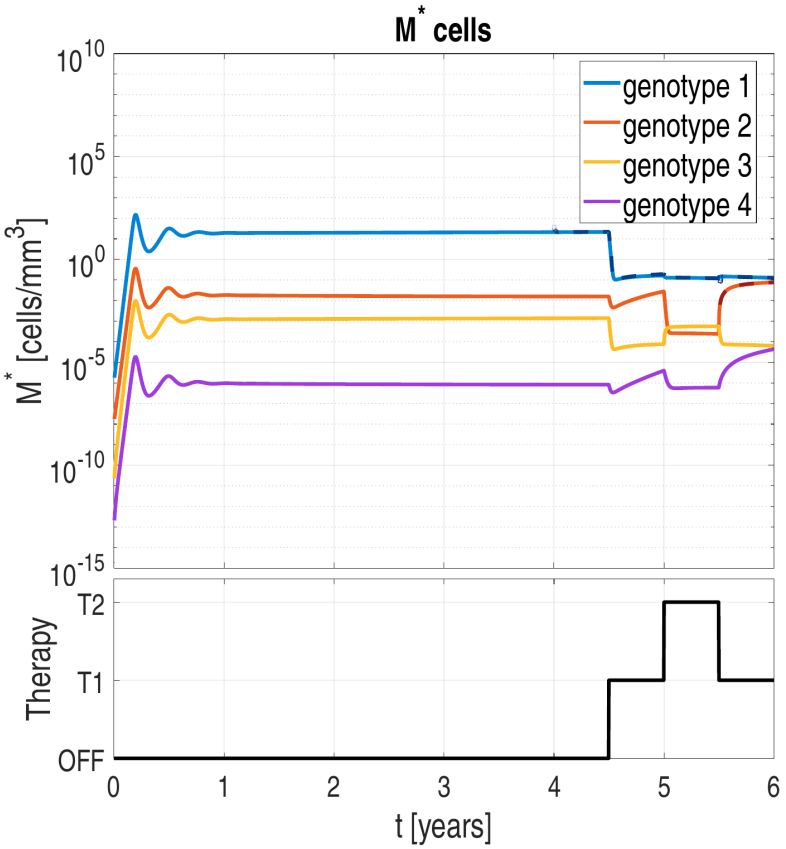


State estimation of infected macrophages, by genotype.

Scenario 2.2: switching therapies



State estimation of infected T cells, by genotype.



State estimation of infected macrophages, by genotype.

Hybrid extended Kalman filtering

Hybrid extended Kalman filtering [7]

1. Consider continuous-time dynamics with discrete-time measurements:

$$\dot{x} = f(x, u, w, t), \quad y_k = h_k(x_k, v_k), \quad w(t) \sim (0, Q), \quad v_k \sim (0, R_k). \tag{5}$$

Hybrid extended Kalman filtering [7]

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2. Initialize the filter:

$$\hat{x}_0^+ = E[x_0], \qquad P_0^+ = E\left[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T\right]. \tag{6}$$

Hybrid extended Kalman filtering [7]

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3. For $k = 1, 2, \ldots$:

   (a) Predict: integrate the estimate and its covariance from $(k-1)^+$ to k^-,

$$\dot{\hat{x}} = f(\hat{x}, u, 0, t), \qquad \dot{P} = AP + PA^T + LQL^T, \tag{7}$$

   with $A = \frac{\partial f}{\partial x}\Big|_{\hat{x}}$, $L = \frac{\partial f}{\partial w}\Big|_{\hat{x}}$, starting from $\hat{x} = \hat{x}_{k-1}^+$, $P = P_{k-1}^+$ and ending at $\hat{x} = \hat{x}_k^-$, $P = P_k^-$. (8)

[7] Simon, D. (2006). **Optimal state estimation: Kalman, H infinity, and nonlinear approaches**. John Wiley & Sons.

Hybrid extended Kalman filtering [7]

3. For $k = 1, 2, \ldots$:

   (b) Update: at time k, incorporate the measurement y_k:

$$
\begin{aligned}
K_k &= P_k^- H_k^T \left(H_k P_k^- H_k^T + M_k R_k M_k^T \right)^{-1}, \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k \left(y_k - h_k(\hat{x}_k^-, 0, t_k) \right), \\
P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k M_k R_k M_k^T K_k^T,
\end{aligned}
\tag{9}
$$

   where H_k, M_k are the partial derivatives of $h_k(x_k, v_k)$ with respect to x_k and v_k, evaluated at $\hat{x}_k^-$.

Hybrid extended Kalman filtering [7]

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$$\begin{aligned} K_k &= P_k^- H_k^T \left(H_k P_k^- H_k^T + M_k R_k M_k^T \right)^{-1}, \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k \left(y_k - h_k(\hat{x}_k^-, 0, t_k) \right), \\ P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k M_k R_k M_k^T K_k^T, \end{aligned} \tag{9}$$

   where H_k, M_k are the partial derivatives of $h_k(x_k, v_k)$ with respect to x_k and v_k, evaluated at $\hat{x}_k^-$.

- Joint state–parameter estimation: to estimate a parameter p, augment the state,

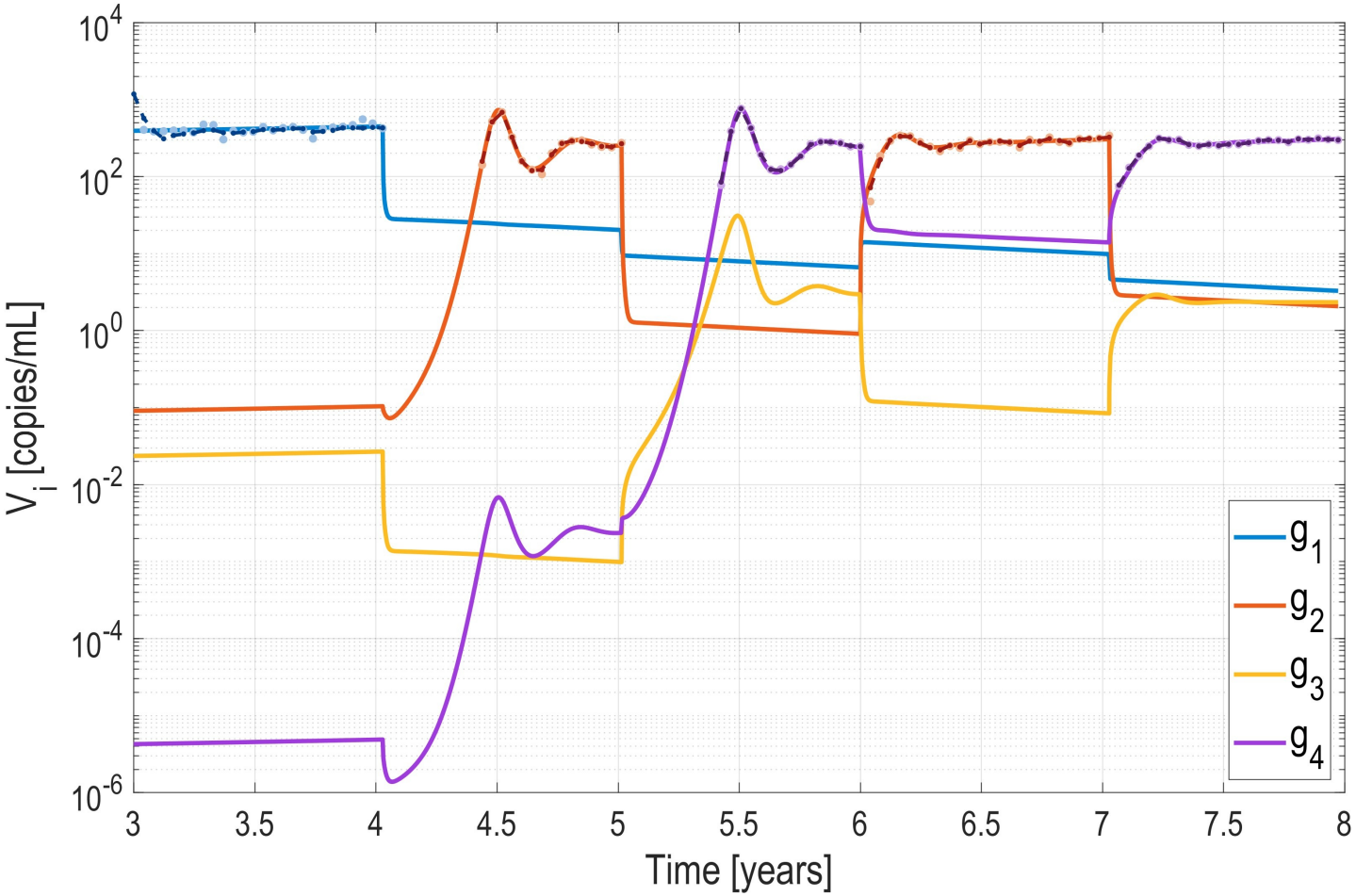
$$x'_k = \begin{bmatrix} x_k \\ p_k \end{bmatrix}, \tag{10}$$

  and if p_k is constant, model $p_{k+1} = p_k + w_{pk}$, where w_{pk} is a small artificial noise that lets the filter change its estimate of p_k.

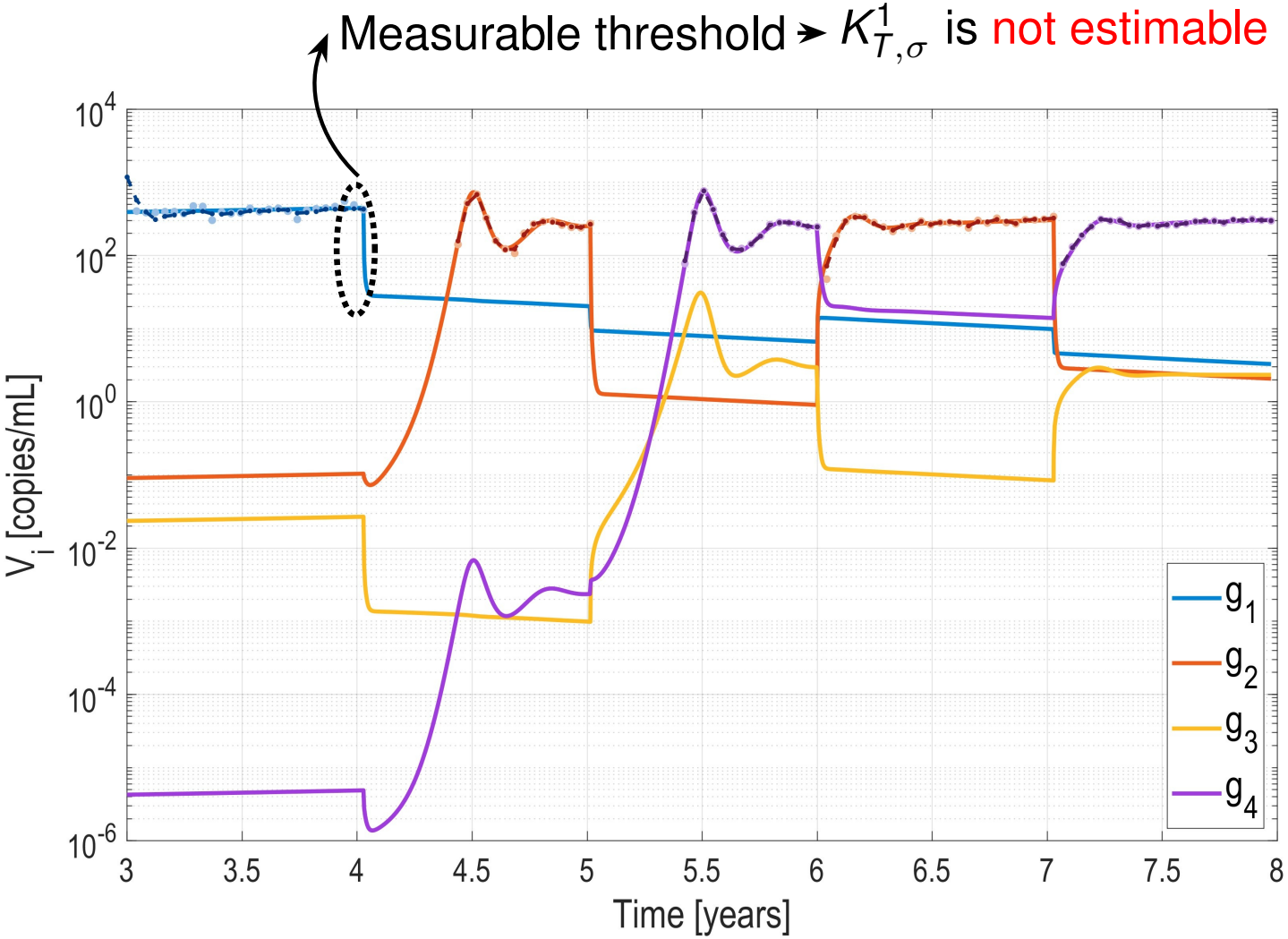
[7] Simon, D. (2006). **Optimal state estimation: Kalman, H infinity, and nonlinear approaches**. John Wiley & Sons.

Results on hybrid EKF

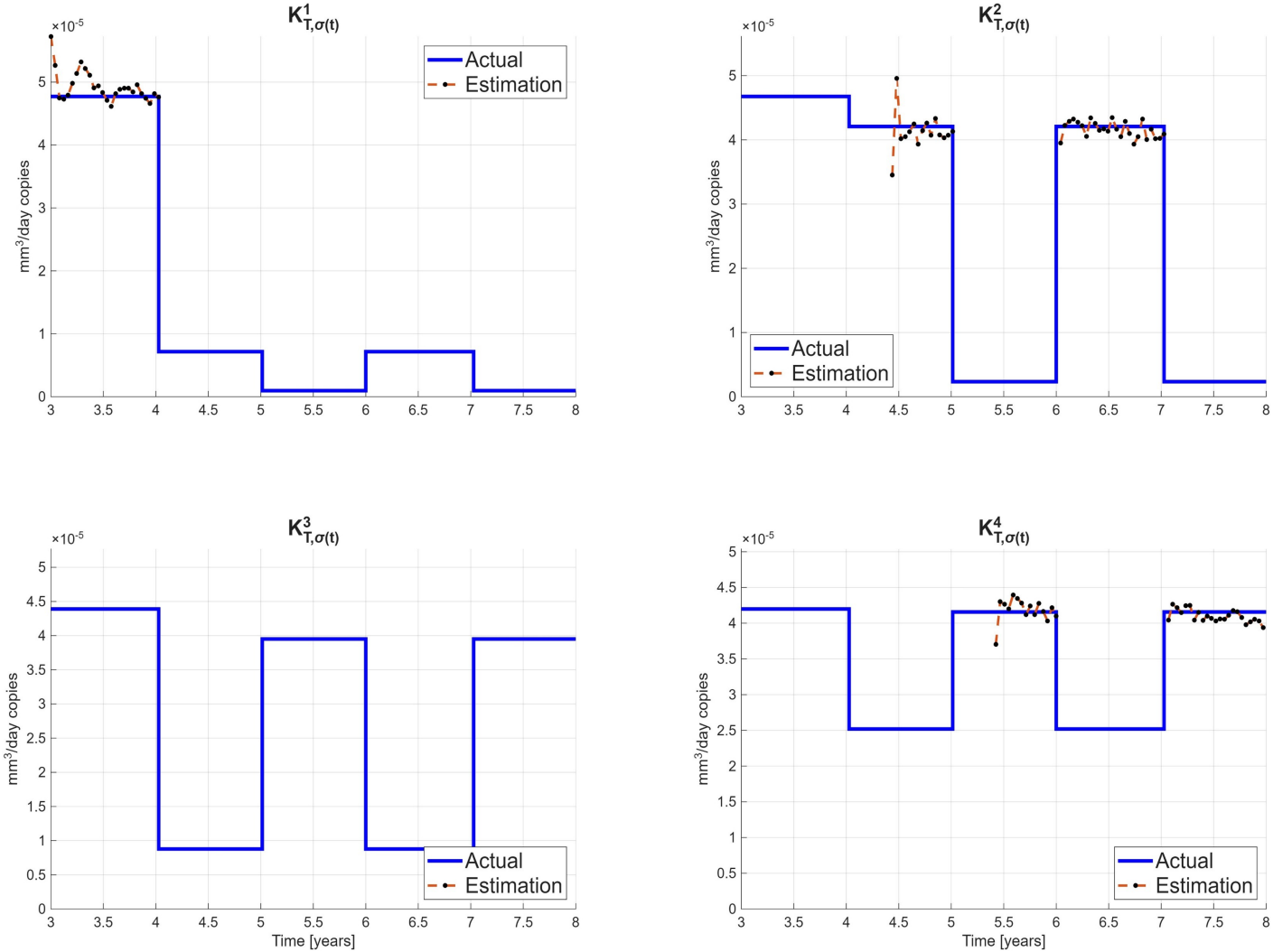
HIV mutation, four genotypes



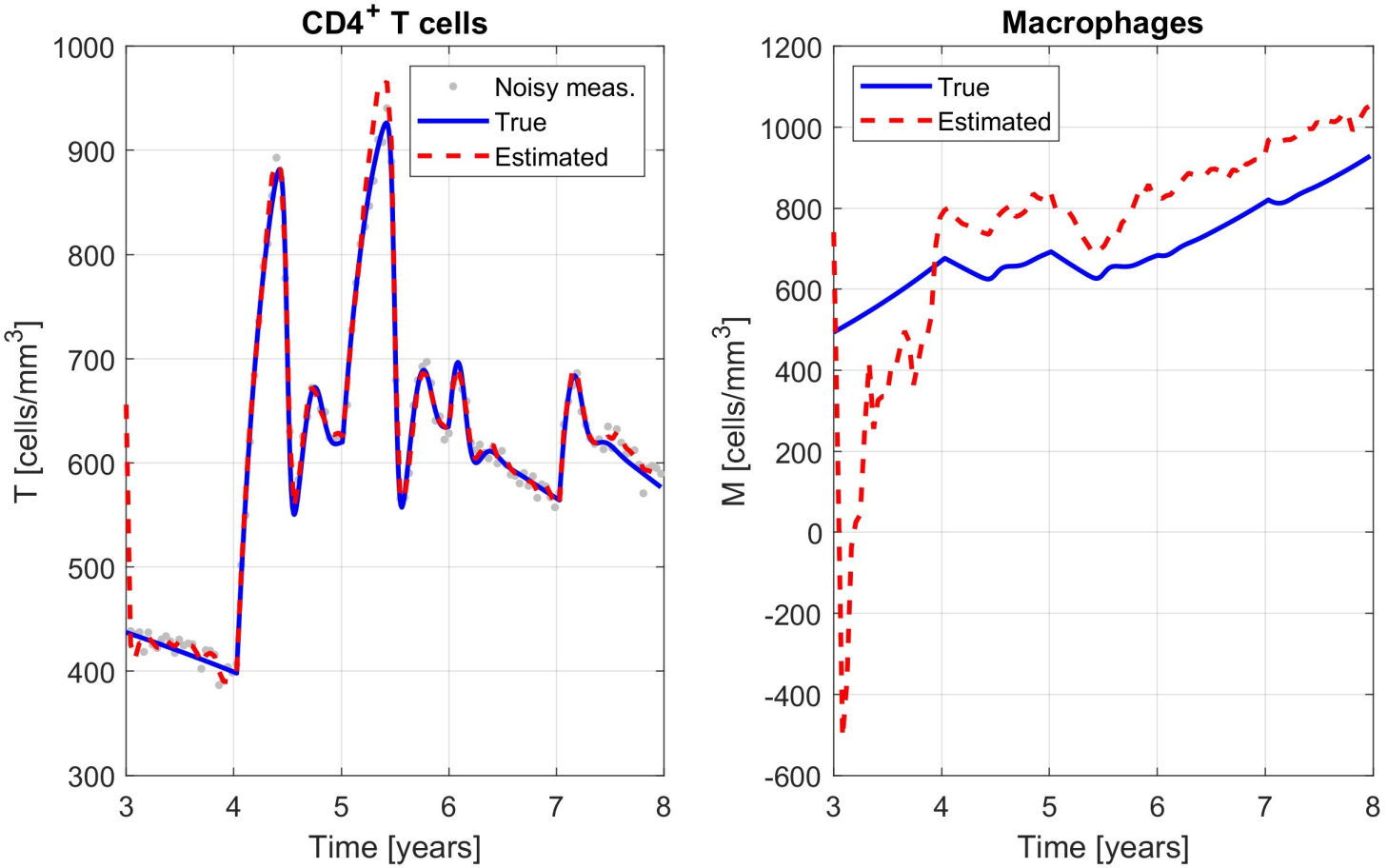
Hybrid EKF: state estimation of each virus genotype.



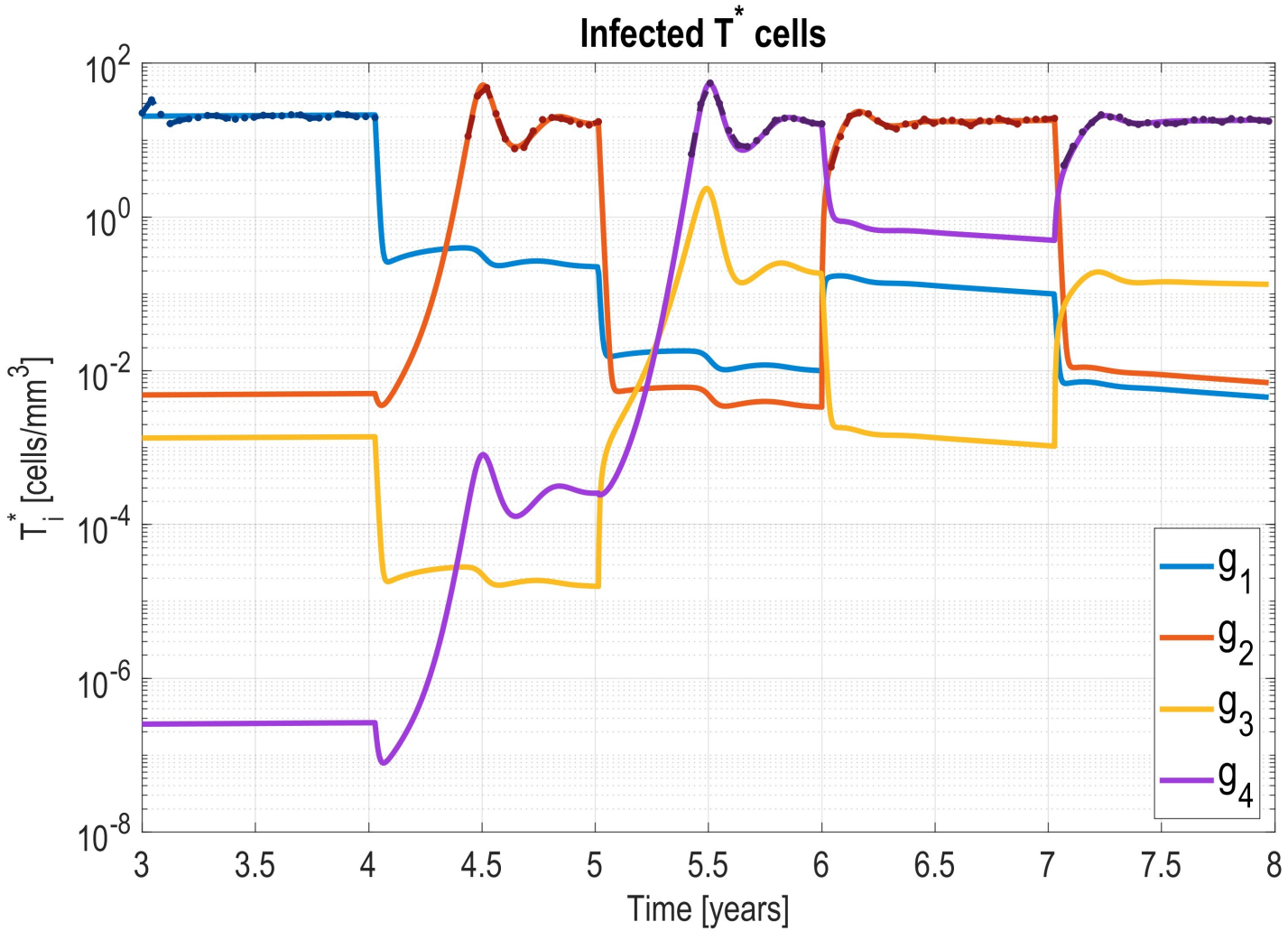
Hybrid EKF: state estimation of each virus genotype.



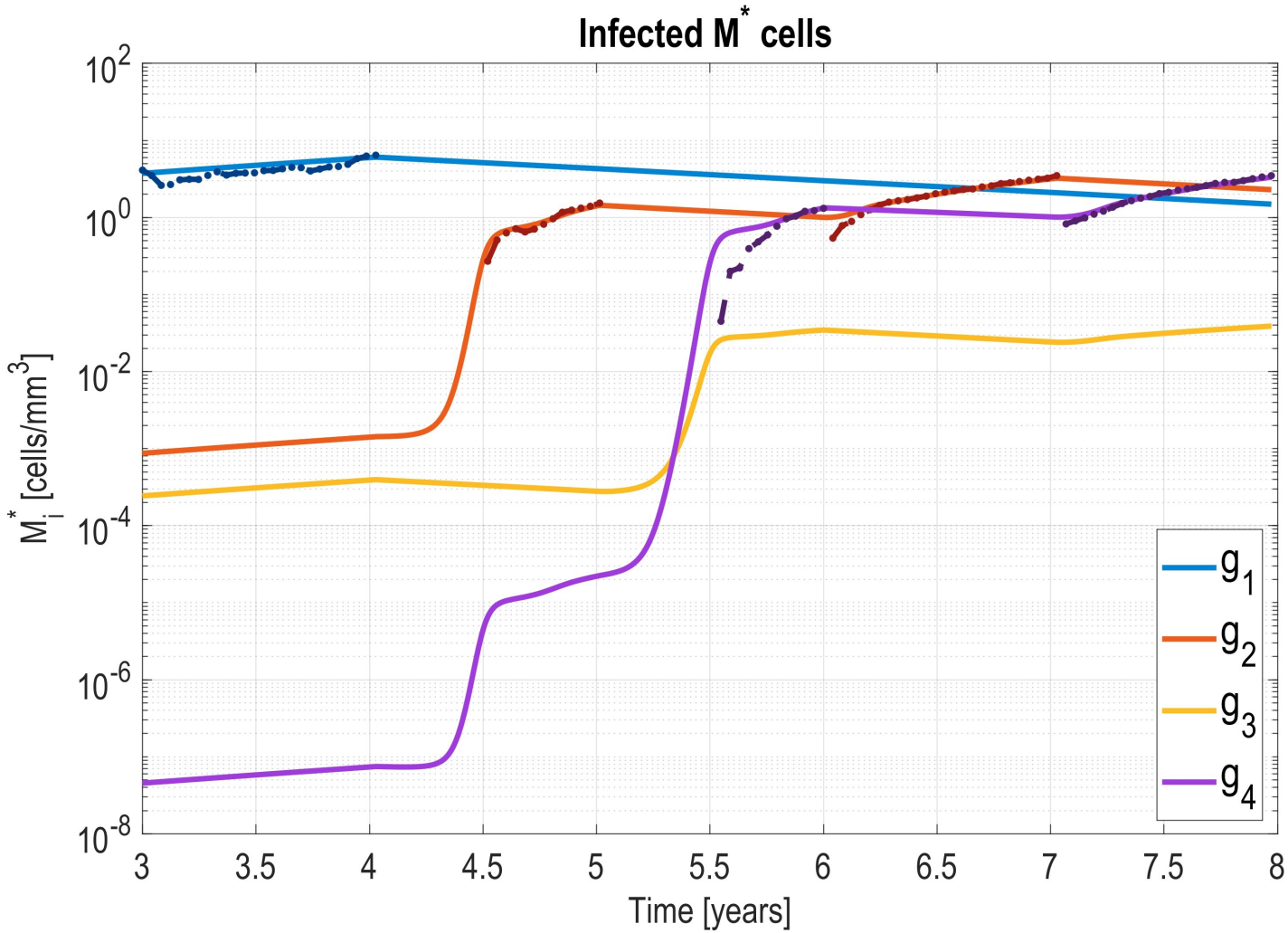
Hybrid EKF: parameter estimation of $K^i_{T,\sigma}$, $i = 1, 2, 3, 4$.



Hybrid EKF: state estimation of $CD4^+$ T cells (left) and macrophages (right).



Hybrid EKF: state estimation of infected T cells.



Hybrid EKF: state estimation of infected macrophages.

Conclusions

- The real-time parameter updating algorithm yields accurate predictions of the HIV infection.
- Adaptive observers are originally formulated in continuous time; their performance therefore suffers under sparse sampling, where the hybrid EKF is a natural alternative.
- On the same HIV model, the hybrid EKF handles noisy, sampled measurements; with both estimators, parameters of genotypes below the measurable threshold are not estimable.
- One promise research direction is the development of Adaptive EKF observers.
- Current proposals in medicine still rely on batch-based identification process and this underpins model's capabilities to predict medical outcomes.

Thanks for listening! Questions?

Contact: jeserenom@gmail.com

https://www.juansereno.github.io

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[9] Hernández-Vargas, E. A., Parra-Rojas, C., & Olaru, S. (2021). Switching logistic maps to design cycling approaches against antimicrobial resistance. In *60th IEEE Conference on Decision and Control (CDC)*, 4248–4253.

[10] Hernandez-Vargas, E. A., Colaneri, P., & Middleton, R. H. (2013). Switching strategies to mitigate HIV mutation. *IEEE Transactions on Control Systems Technology*, 22(4), 1623–1628.

Backup: performance results

- Root mean squared logarithmic error (RMSLE) of the adaptive observer.[†]

Scenario 1.1

	$K^1_{T,\sigma}$	Virus	CD4$^+$T	Mac.	Inf. T	Inf. Mac.
RMSLE	1.662×10^{-6}	0.1834	0.0109	0.0520	0.0447	0.0021

Scenario 1.2

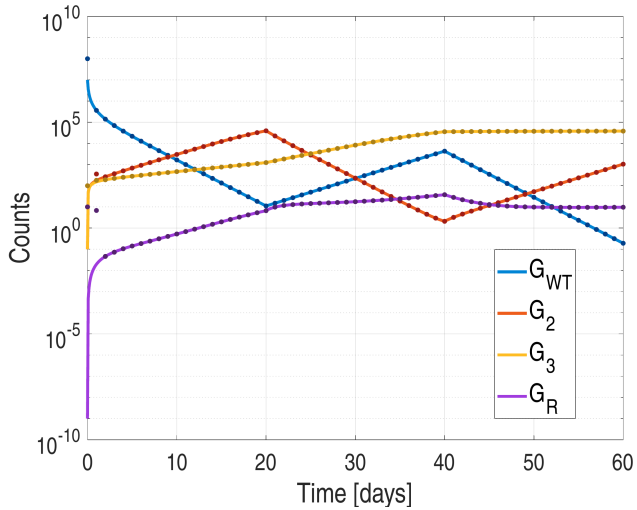
	$K^1_{T,\sigma}$	Virus	CD4$^+$T	Mac.	Inf. T	Inf. Mac.
RMSLE	1.022×10^{-7}	0.0030	1.792×10^{-4}	0.0081	6.243×10^{-4}	6.327×10^{-4}

[†] Both scenarios consider $K_{T,\sigma}$ as an unknown parameter, and only V and T as measurable variables.

Backup: beyond HIV — antimicrobial resistance [9]

- Recall the opening case: *K. pneumoniae* and antibiotic failure.
- The same hybrid EKF applies to a bacterial model with antibiotic-dependent (switching) growth rates:

$$\dot{x}_i = \rho_{i,\sigma(t)}\, x_i\left(1 - \frac{x_i}{K}\right) - \delta x_i + \mu \sum_{j \neq i} m_{i,j}\, x_j.$$



Hybrid EKF on the AMR model: bacterial populations by genotype.

[9] Hernández-Vargas, E. A., Parra-Rojas, C., & Olaru, S. (2021). **Switching logistic maps to design cycling approaches against antimicrobial resistance**. *60th IEEE CDC*, 4248–4253.

Backup: an antimicrobial resistance model [9]

- The population x_i of each bacterial genotype evolves as

$$\dot{x}_i(t) = \rho_{i,\sigma(t)}\, x_i(t)\left(1 - \frac{x_i(t)}{K}\right) - \delta x_i(t) + \mu \sum_{j \neq i}^{\eta} m_{i,j}\, x_j(t), \quad (4)$$

where the growth rates $\rho_{i,\sigma(t)}$ switch with the antibiotic $\sigma(t)$ being applied.

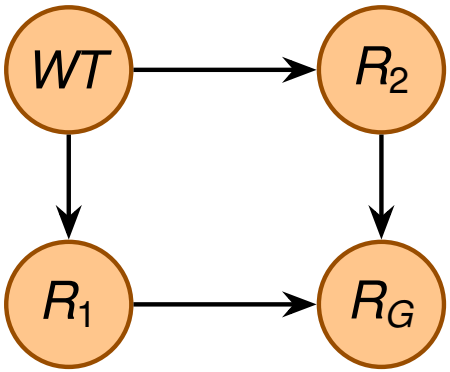
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  where the growth rates $\rho_{i,\sigma(t)}$ switch with the antibiotic $\sigma(t)$ being applied.

- Mutations follow the mutation tree:



[9] Hernández-Vargas, E. A., Parra-Rojas, C., & Olaru, S. (2021). **Switching logistic maps to design cycling approaches against antimicrobial resistance**. *60th IEEE CDC*, 4248–4253.

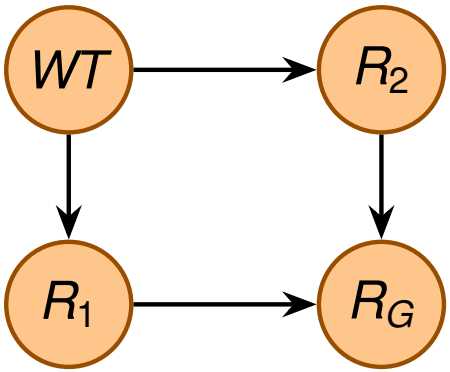
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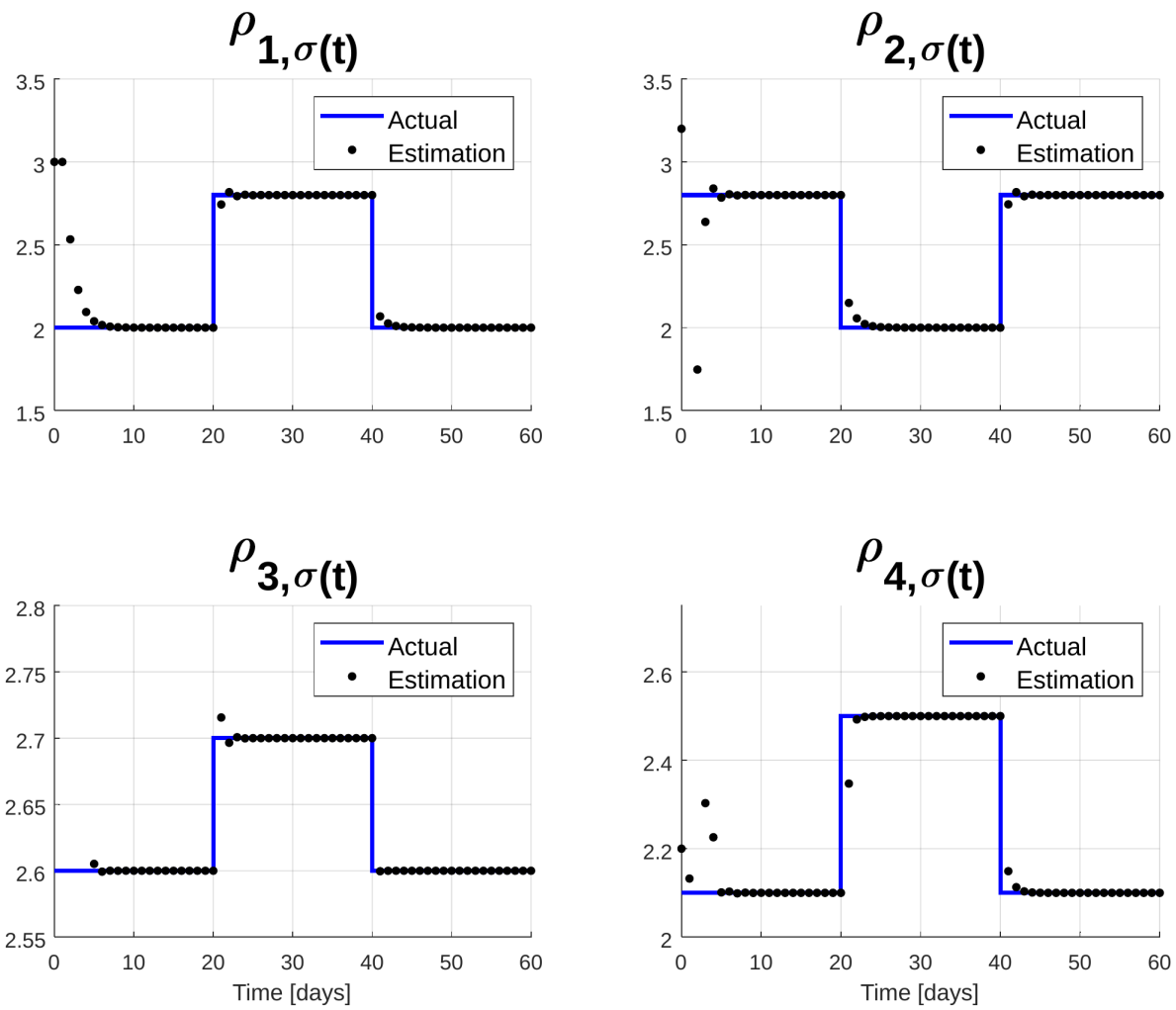
- Mutations follow the mutation tree:



$$M_u = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

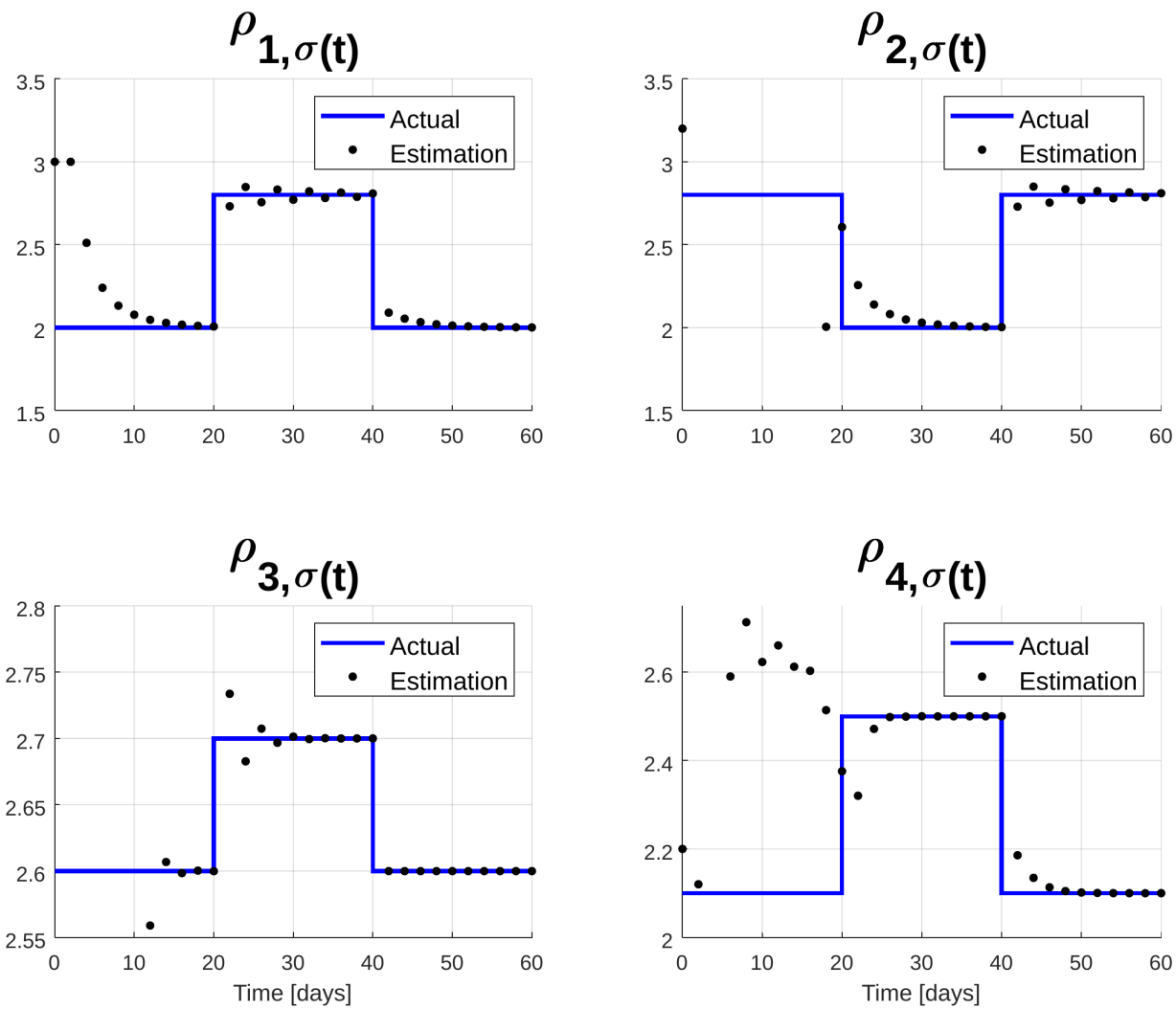
[9] Hernández-Vargas, E. A., Parra-Rojas, C., & Olaru, S. (2021). **Switching logistic maps to design cycling approaches against antimicrobial resistance**. *60th IEEE CDC*, 4248–4253.

Backup: hybrid EKF on the AMR model



Hybrid EKF on the AMR model: parameter estimation of $\rho_{i,\sigma}$.

Backup: hybrid EKF on the AMR model



Parameter estimation of $\rho_{i,\sigma}$ with sampling every 2 days.